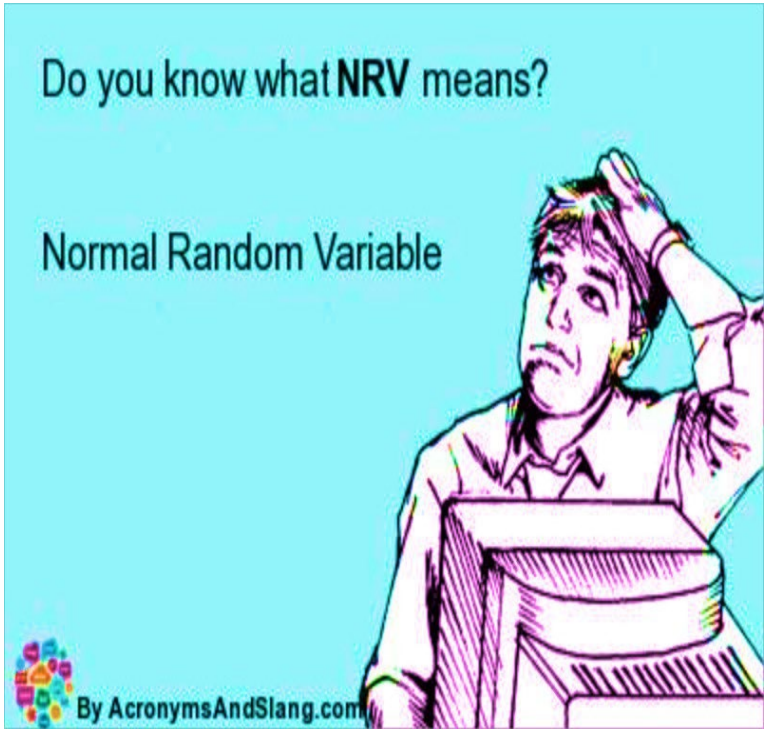
So you have fitted a model, then what?

More questions to ask:

* Does the model help (or do we need to bother)?
* What can we say about the regression coefficients?
* Suppose that DAX change by 1%, how should we **expect** the DJ to change on average? How do we **predict** the actual change of DJ on a **particular** day?
* Does the fitted model fit the data well? Are any model assumptions being violated?
* If there is a problem, what do we do?

Facts about Normal Distribution

* A constant plus a normal random variable yields a normal random variable
* Any linear combination (e.g., sums and differences) of several normal variables yields a normal random variable



Y is Normal Distributed

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Since we assumed ε_i is normal, then constant $+\varepsilon_i$ is also normal

$$Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$$

What about b_1 ?

* b_1 is a linear combination of the Y_i's, i.e.,

$$\boldsymbol{b_1 = k_1 Y_1 + \cdots + k_n Y_n \; for \; some \; k_1, \ldots, k_n}$$

* *A linear combination of normally distributed random variables is normally distributed*
* Then $\boldsymbol{b_1 \sim N(\mu_{b_1}, \sigma^2_{b_1})}$. How do we find $\boldsymbol{\mu_{b_1}}$ and $\boldsymbol{\sigma^2_{b_1}}$?
* b_1 is an unbiased estimator of $\boldsymbol{\beta_1}$, thus, $\boldsymbol{\mu_{b_1} = \beta_1}$.

Sampling distribution of b_1

* The sampling distribution of b_1 is normal with $mean\ \beta_1$ and $variance\ \frac{\sigma^2}{\sum(X_i-\bar{X})^2}$
* $\boldsymbol{\sigma^2}$ is estimated by *MSE*, hence $\boldsymbol{\sigma^2_{b_1}}$ is estimated by

$$s(b_1)^2 = \frac{MSE}{\sum(X_i - \bar{X})^2}$$

* SSE/σ^2 is χ^2 with (n-2) degrees of freedom and is independent of b_1
* $t = (b_1 - \beta_1)/s(b_1)$ follows the ***t – distribution with n-2 degrees of freedom.***

Confidence interval for β_1

* *Let* $t_{\alpha/2,n-2}$ *be the t-value that cuts off α/2 area to its left for a t distribution with (n-2) degrees of freedom.*
* *Formula for a (1-α) 100% CI for estimating* β_1 *is:*

$$b_1 \pm t_{\alpha/2,n-2}\, s(b_1)$$

* ***Note:*** *Probability Statement is regarding the interval - not parameter since the parameter is fixed (though it is unknown)*

One-sided Hypothesis tests for β_1

$H_0: \beta_1 = \beta_1^*$

$H_1: \beta_1 > \beta_1^*$ *or* $\beta_1 < \beta_1^*$

$$t = \frac{b_1 - \beta_1^*}{s(b_1)}$$

Assume H_0 is true

Gather sample evidence

Find probability of repeating the experiment and finding

More extreme evidence in support of H1

Inference for β_0

* b_0 is a linear combination of the Y_i's which are normally distributed; therefore it is normally distributed.

$$\boldsymbol{t = \frac{b_0 - \beta_0}{s(b_0)} \sim t_{n-2}}$$

* Inference for β_0 can be done similarly to that for β_1.

Example: How are the different stock indices related?

* Test whether β_1 is different from zero. Write out the null and alternative hypotheses.
* Obtain the t statistic value.
* Verify that this value was calculated correctly.
* What is the p-value?
* Obtain 95% confidence intervals for β_0 and β_1. Interpret your results.

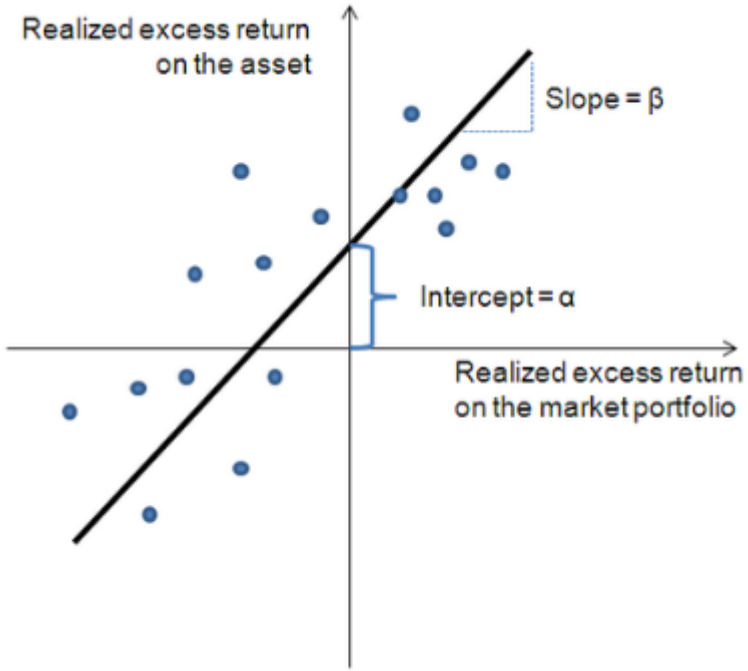
The Security Characteristic Line

* **Security characteristic line** (SCL) is a regression line, plotting performance of a particular security or portfolio against that of the market portfolio at every point in time. The SCL is plotted on a graph where the Y-axis is the excess return on a security over the risk-free return (e.g., the return on Treasury Bills) and the X-axis is the excess return of the market in general. The slope of the SCL is the security's beta, and the intercept is its alpha (from Wikipedia)

* The formula is

$$R_{i,t} - R_f = \alpha_i + \beta_i\left(R_{M,t} - R_f\right) + \epsilon_{i,t}$$

  * $R_{i,t}$ and $R_{M,t}$ are the returns of the investment and the market (e.g., S&P 500) at time t
  * R_f is the return of a risk-free return (e.g., the return on Treasury Bills)
  * α_i and β_i are the asset's alpha and beta coefficients



The Alpha and Beta Coefficients

* **Alpha** is a measure of the active return on an investment, the performance of that investment compared to a suitable market index.
  * A positive (negative) alpha means the investment outperformed (underperformed) the market.
* **Beta** of an investment indicates whether the investment is more or less volatile than the market.
  * A Beta more than 1 indicates that the investment is more volatile than the market.
  * Volatility is measured as the fluctuation of the price around the mean: the standard deviation.

Relations to the regression coefficients in simple linear regression

* Our regression analysis essentially considered

$$R_{i,t} = \beta_0 + \beta_1 R_{M,t} + \epsilon_{i,t}$$

* So we ignored R_f, i.e., the return of a risk-free return
* Since R_f is expected to vary little over time (or can be treated as a constant since it's risk-free), β_1 is the β coefficient in the security characteristic line.
* β_0 is NOT the α coefficent

Example: How are the different stock indices related?

* Formulate the right hypotheses to test whether DJ is more volatile than DAX.

* Calculate the test statistic

* Find the p-value. What is your conclusion?

So you have fitted a model, then what?

More questions to ask:

* Does the model help (or do we need to bother)?
* What can we say about the regression coefficients?
* Suppose that DAX change by 1%, how should we expect the DJ to change on average? How do we predict the actual change of DJ on a particular day?
* Does the fitted model fit the data well? Are any model assumptions being violated?
* If there is a problem, what do we do?

Estimating mean of Y for a given X

Estimating $E(Y_h)$

-- *h is any level of X that may or may not be one of the original n levels.*

$$E(Y_h) = \beta_0 + \beta_1 X_h,$$

$$\textit{Estimate: } \hat{Y}_h = b_0 + b_1 X_h$$

Recall that both b_0 and b_1 are linear in Y_i's, so is $\hat{Y}_h$, i.e.,

$$\hat{Y}_h = \sum_{i=1}^{n} k_{h,i} Y_i$$

for some $k_{h,i}$'s.

Inference about E(Y)

* The variance of $\hat{Y}_h$ is

$$\boldsymbol{Var(\hat{Y}_h) = \sum_{i=1}^{n} (k_{h,i})^2 \sigma^2}$$

* How do you estimate the variance of $\hat{Y}_h$?
* Let $s(\hat{Y}_h)$ be the standard error estimate of $\hat{Y}_h$.

$$\boldsymbol{\frac{\hat{Y}_h - E(Y_h)}{s(\hat{Y}_h)} \sim t_{n-2}}$$

* This statistic can then be used to form a confidence interval for estimating $E(Y_h)$.
* Since

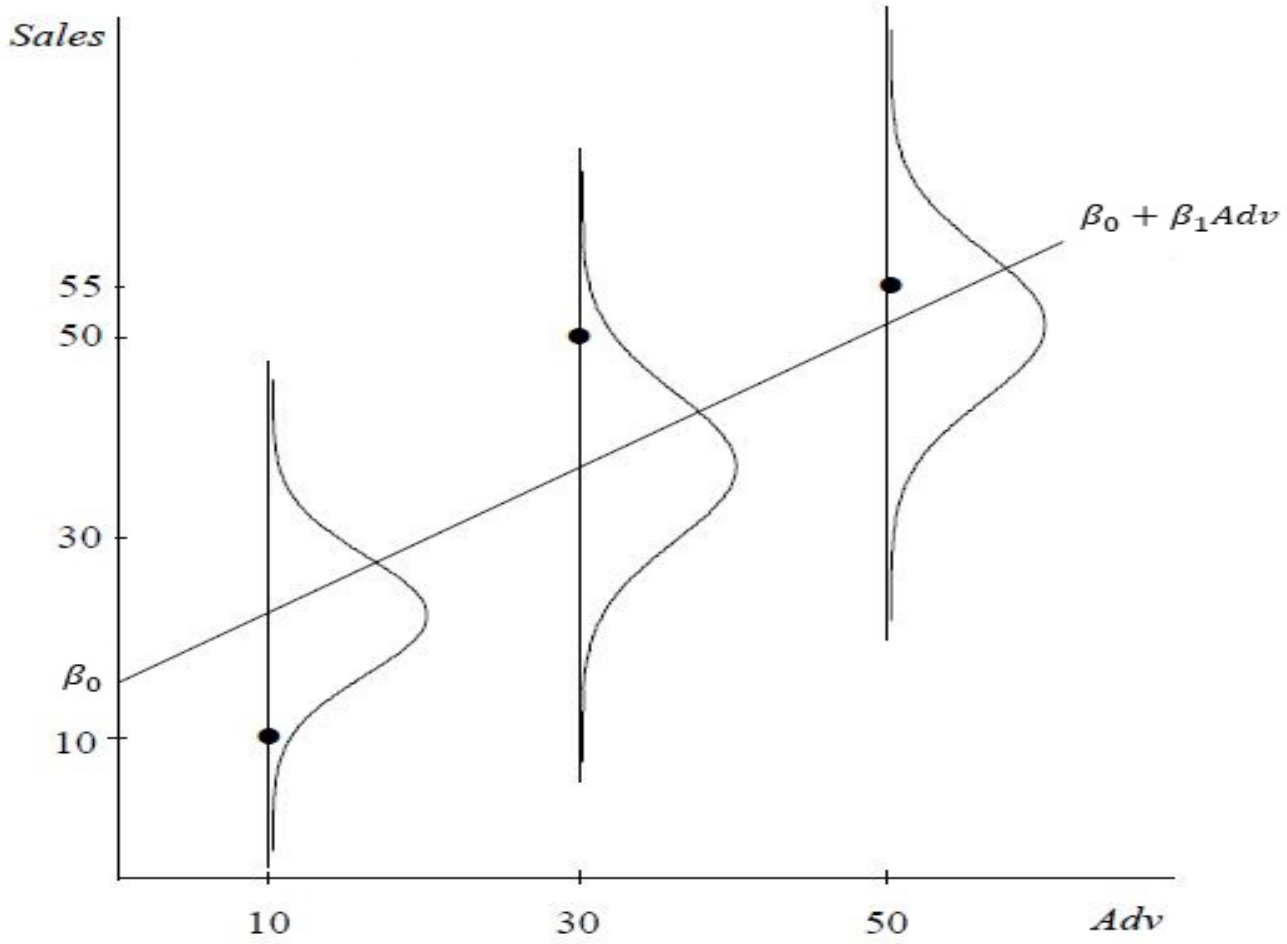
$$P\left(-t_{\frac{\alpha}{2},n-2} \leq \frac{\hat{Y}_h - E(Y_h)}{s(\hat{Y}_h)} \leq t_{\frac{\alpha}{2},n-2}\right) = 1-\alpha$$

* We have

$$P\left(\hat{Y}_h - t_{\frac{\alpha}{2},n-2}s(\hat{Y}_h) \leq E(Y_h) \leq \hat{Y}_h + t_{\frac{\alpha}{2},n-2}s(\hat{Y}_h)\right) = 1-\alpha$$

* A 100(1-α) % CI for estimating $E(Y_h)$:

$$\hat{Y}_h \pm t_{\alpha/2,n-2}s(\hat{Y}_h)$$


Sales
$\beta_0 + \beta_1 Adv$
55
50
30
β_0
10
10
30
50
Adv

Prediction of a New Observation

* A new observation at X_h: $Y_{h(new)}$
* The point prediction of $Y_{h(new)}$ is $\boldsymbol{\hat{Y}_h = b_0 + b_1 X_h}$
* Under the normal error regression model, $\hat{Y}_h - Y_{h(new)}$ is normal.
* If $P\left(-t_{\frac{\alpha}{2},n-2} \leq \frac{\hat{Y}_h - Y_{h(new)}}{\Delta} \leq t_{\frac{\alpha}{2},n-2}\right) = 1 - \alpha$ for some Δ, then

  $P\left(\hat{Y}_h - t_{\frac{\alpha}{2},n-2}\Delta \leq Y_{h(new)} \leq \hat{Y}_h + t_{\frac{\alpha}{2},n-2}\Delta\right) = 1 - \alpha.$
* Prediction interval: $\hat{Y}_h \pm t_{\alpha/2,n-2}\Delta$. But what is Δ?
* When will $\frac{\hat{Y}_h - Y_{h(new)}}{\Delta}$ be a t random variable with $df = n - 2$?

$$Var(\hat{Y}_h - Y_{h(new)}) = Var(\hat{Y}_h) + Var(Y_{h(new)}) = Var(\hat{Y}_h) + \sigma^2$$

* The variance needs to accounts for:
  * *Variability in estimating the sample regression line, $\hat{Y}_h$*
  * *Variability in individual observations about the regression line ($= \sigma^2$).*
* How do you estimate the variance of $\hat{Y}_h - Y_{h(new)}$?

* Set $\Delta = s(\hat{Y}_h - Y_{h(new)})$, the standard error estimate of $s(\hat{Y}_h - Y_{h(new)})$.
* Note that $s(\hat{Y}_h - Y_{h(new)})$ is much bigger than $s(\hat{Y}_h)$.

* A (1-α)100% prediction interval is

$$\widehat{Y}_h \pm t_{\frac{\alpha}{2},n-2} s(\widehat{Y}_h - Y_{h(new)})$$

* We are (1-α)100% confident that this interval captures the **individual** value of Y_h at $X = X_h$ inside.

* *The prediction interval is wider than the confidence interval.*

Confidence Interval or Prediction Interval?

* What will the humidity level in the classroom **tomorrow** when we set the temperature at 25 °C?
* How much do families whose disposable income is $80,000 spend, **on the average**, for meals away from home?
* How much electricity will be consumed **next month** by commercial and industrial users in Shenzhen, given that the business activity for the area remains at the present level?

Example: How are the different stock indices related?

* Suppose that DAX changes by 1%, obtain a 95% confidence interval for the expected changes in DJ, i.e., the average changes in DJ on all such days when DAX changes by 1%.

* Suppose that for one day the DAX changes by 1%, obtain a prediction interval for the actual change of DJ on that particular day.

* How would interpret these results differently? Can you think of more examples when you may need confidence intervals and when you may need prediction intervals?

So you have fitted a model, then what?

More questions to ask:

* Does the model help (or do we need to bother)?
* What can we say about the regression coefficients?
* Suppose that DAX change by 1%, how should we expect the DJ to change on average? How do we predict the actual change of DJ on a particular day?
* Does the fitted model fit the data well? Are any model assumptions being violated?
* If there is a problem, what do we do?

Regression Diagnostics

Help to determine if a model is appropriate for the data. (i.e., are the assumptions reasonable):

1. *Are all the assumptions satisfied?*
2. *Are there any problematic points?*
3. *What are the remedial procedures if needed?*

Residual Analysis

Useful in verifying if the model is appropriate.

- *Residuals do not estimate the model errors, but the behavior of the residuals can give us important information about the validity of the regression assumptions.*

Types of residuals

* **1. REGULAR RESIDUALS:** $e_i = Y_i - \hat{Y}_i$
* **2. STANDARDIZED RESIDUALS** = The residual divided by an estimate of its standard error which varies from case to case, depending on the distance of each case's values of the independent variables from the means of the independent variables.

$$\frac{Y_i - \widehat{Y_i}}{s(e_i)} = \frac{e_i}{s(e_i)} \quad s(e_i) \text{ is a function of MSE}$$

- ## 3. (STUDENTIZED) DELETED RESIDUALS:

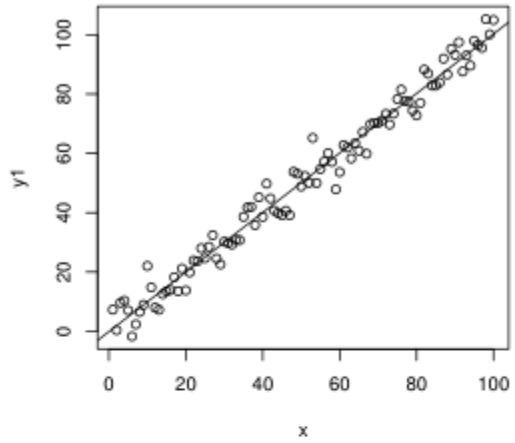
Deleted residuals $\boldsymbol{d_i = Y_i - \hat{Y}_{i(i)}}$ The residual for a case when that case is excluded from the calculation of the regression coefficients. It is the difference between the value of the dependent variable and the adjusted predicted value.

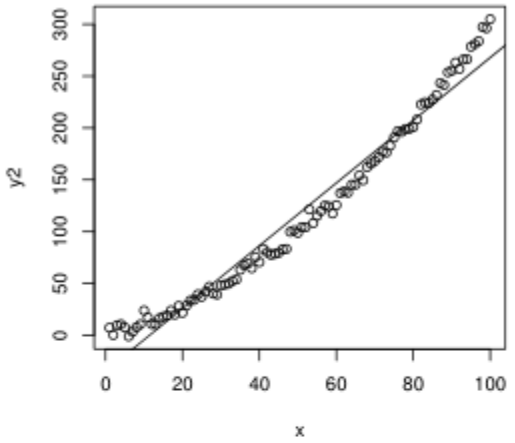
Studentized deleted residuals $\frac{\boldsymbol{d_i}}{\boldsymbol{s(d_i)}}$, where $s(d_i)$ is a function of $MSE_{(i)}$ which is the MSE computed with the ith case deleted.

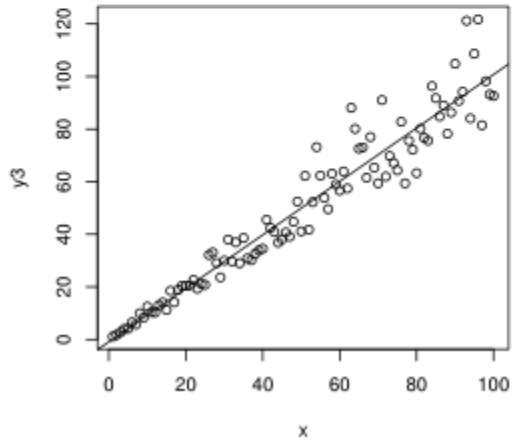
Residuals diagnostics can suggest

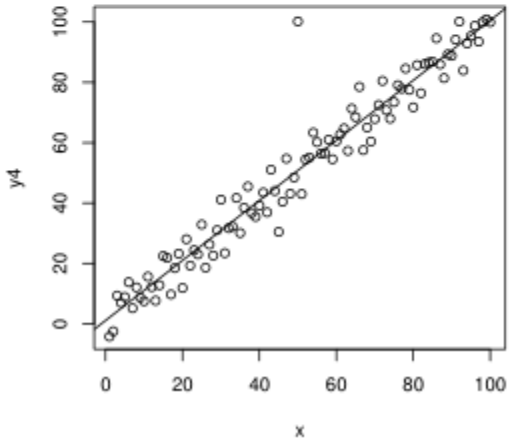
- if the regression model is linear (use Residuals vs Fitted, together with scatter plot)
- if the error terms have constant variance (use Residuals vs Fitted and/or Scale-Location)
- if the error terms are independent (plot residuals against time)
- if there are outliers present (use Residuals vs Fitted and/or Residuals vs Leverage)
- if the error terms are normally distributed (use Normal Q-Q plot)
- if one or more important variables is omitted from the model (use Residuals vs omitted predictor)

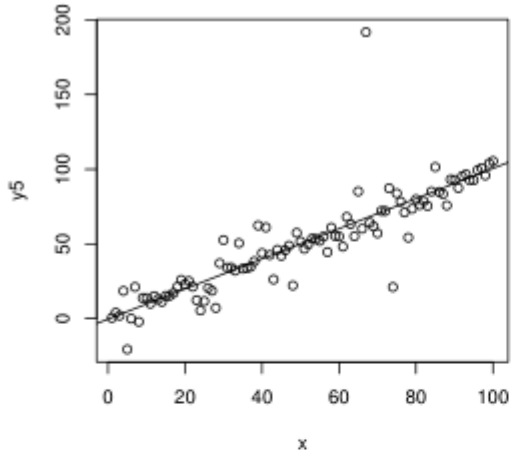
Some Examples

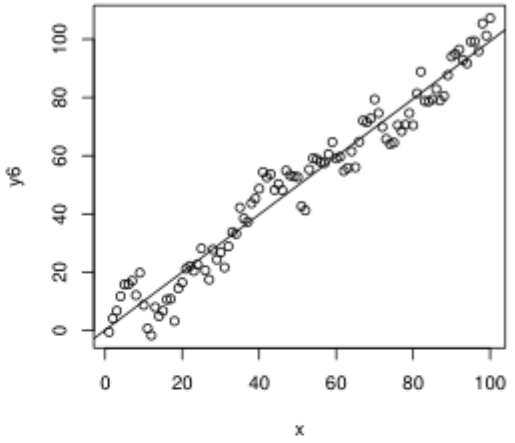


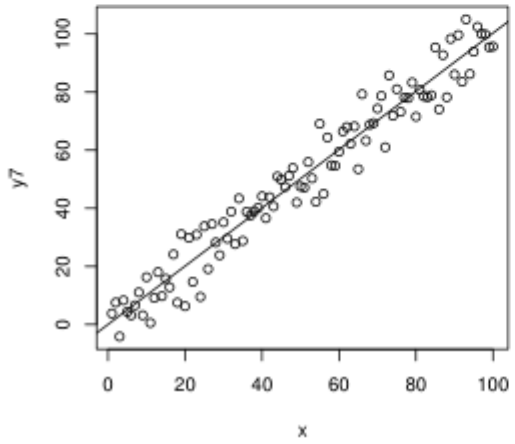


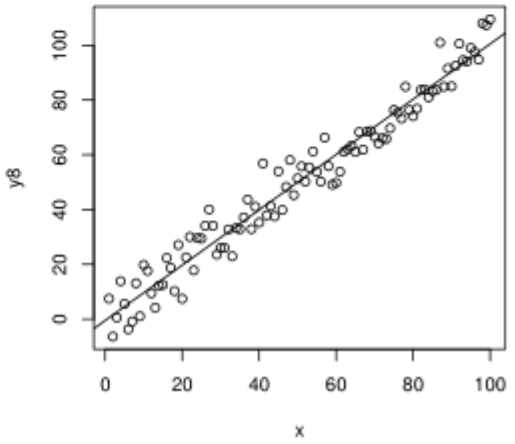




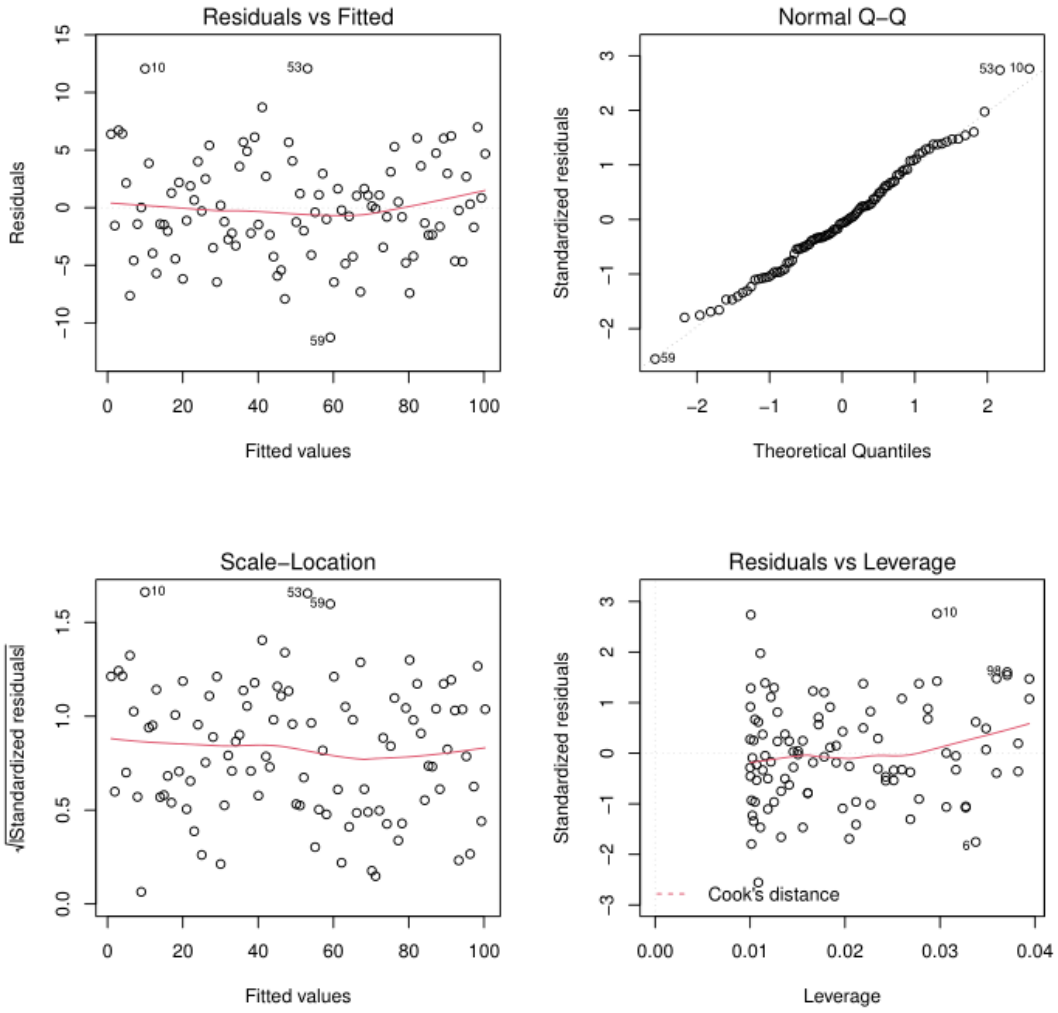

y5
x
0
50
100
150
200
0
20
40
60
80
100


y6
x
0
20
40
60
80
100
0
20
40
60
80
100


y7
x
0
20
40
60
80
100
0
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y8
x
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40
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80
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0
20
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80
100

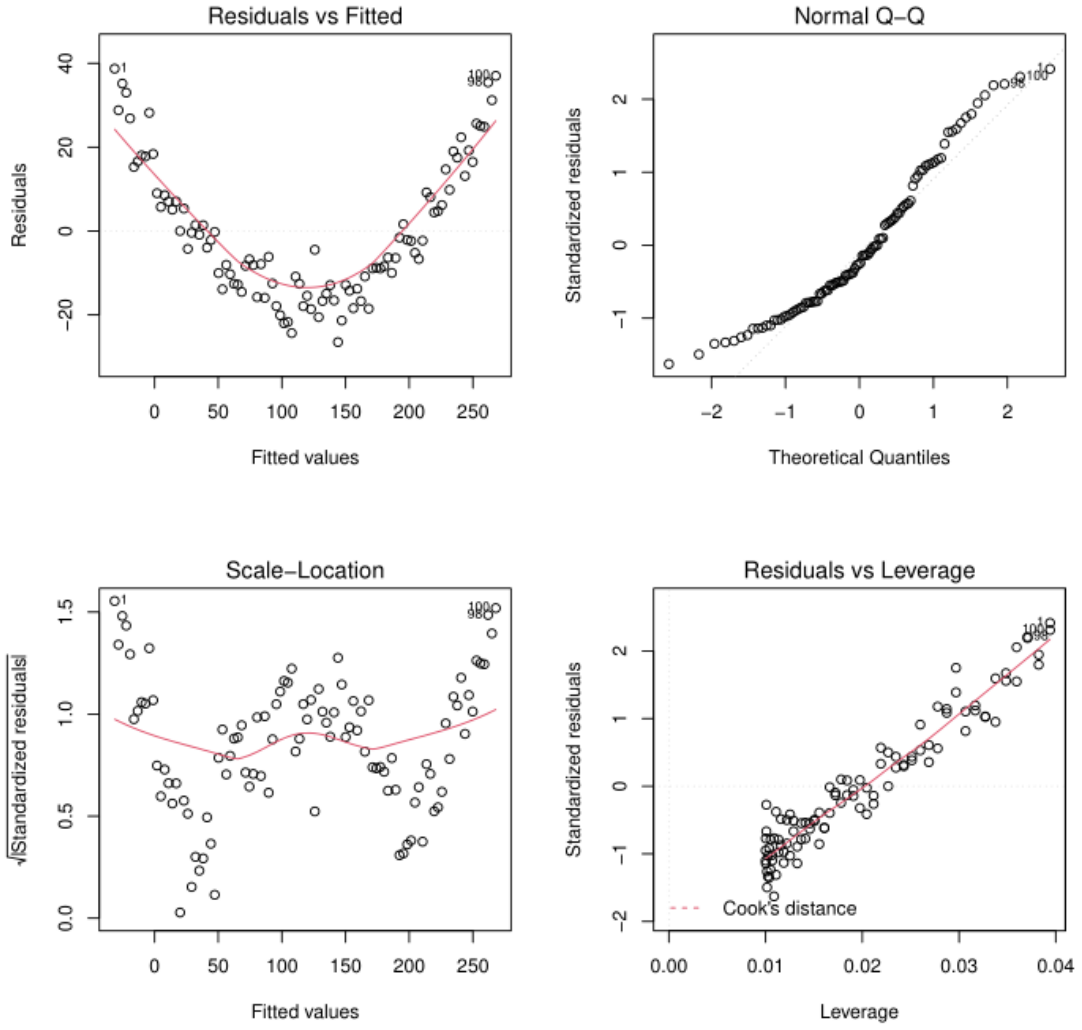
Residual plots for (x,y1)



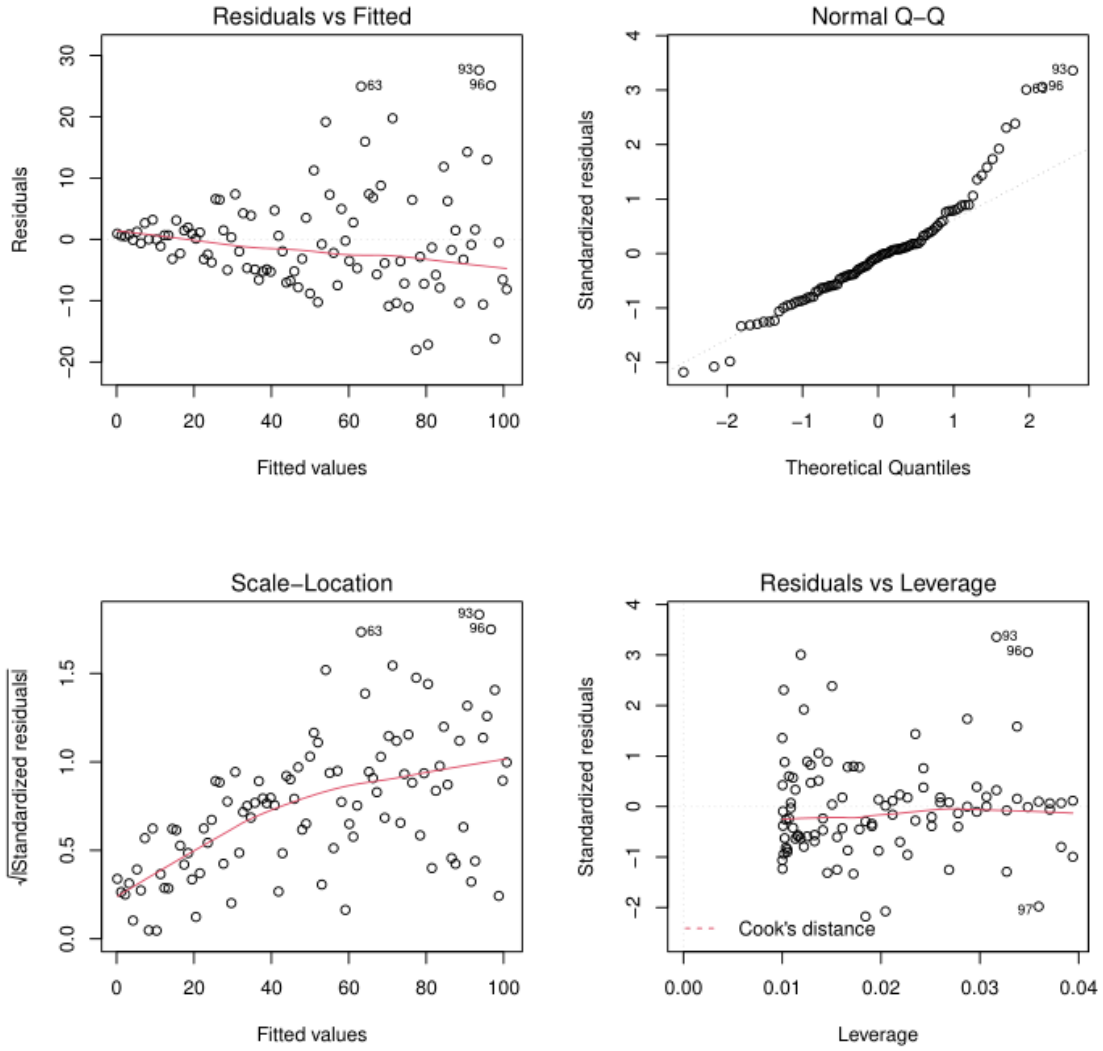
How to read these plots?

- Residuals vs Fitted: If the average residual values are close to 0 for all different fitted values, then linear relationship is good.
- Normal Q-Q plot: The errors are considered normal if the residuals follow the straight dashed line closely.
- Scale-Location: Horizontal line with equally spread points is a good indication of equal variance (homoscedasticity).
- Residuals vs Leverage: Large absolute residuals are extreme values in Y, and large leverages are extreme values in X. Used to identify influential cases, that is, extreme values that might influence the regression results when included or excluded from the analysis. This plot will be described in details later.

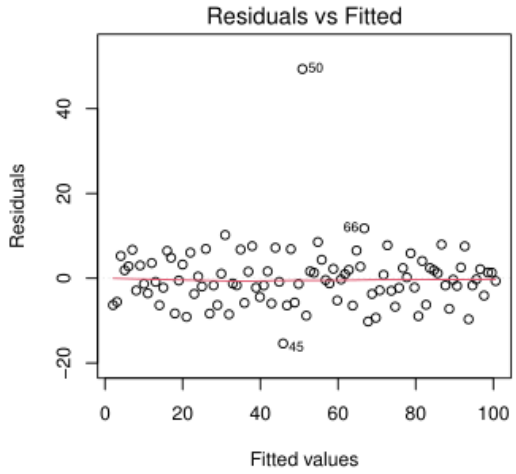
Residual plots for (x,y2)

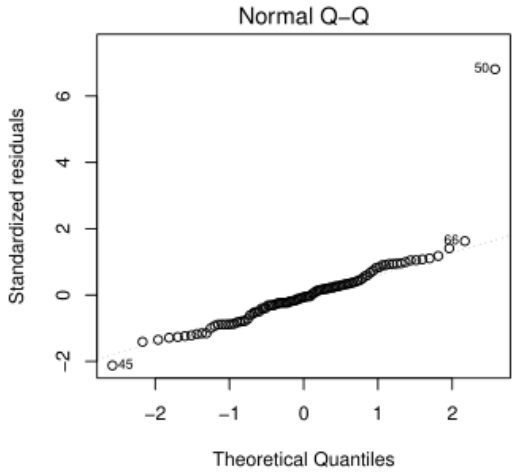


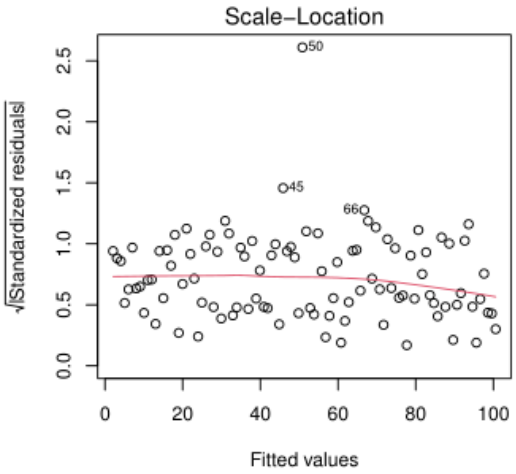
Residual plots for (x,y3)

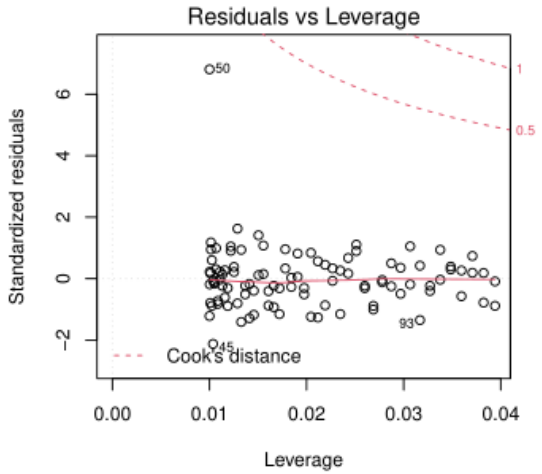


Residual plots for (x,y4)

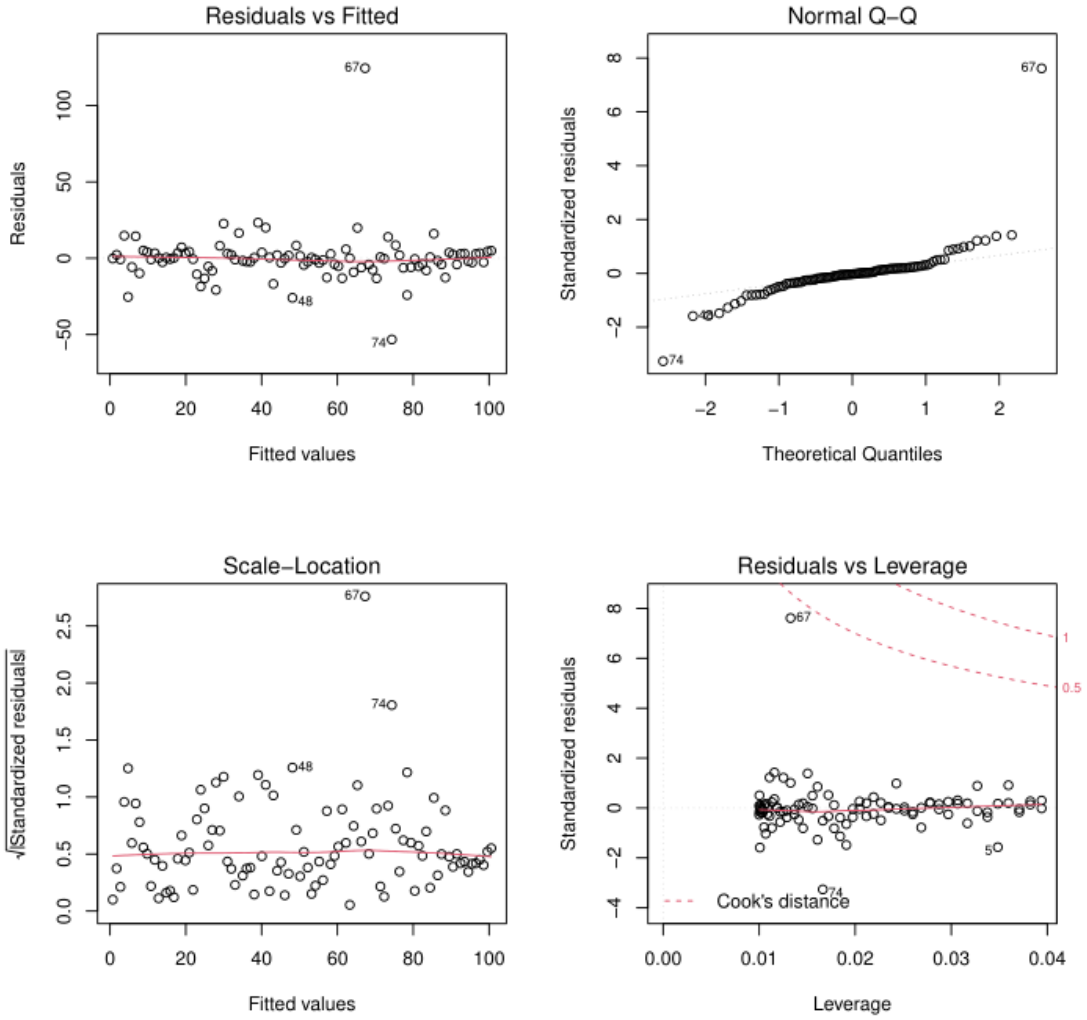




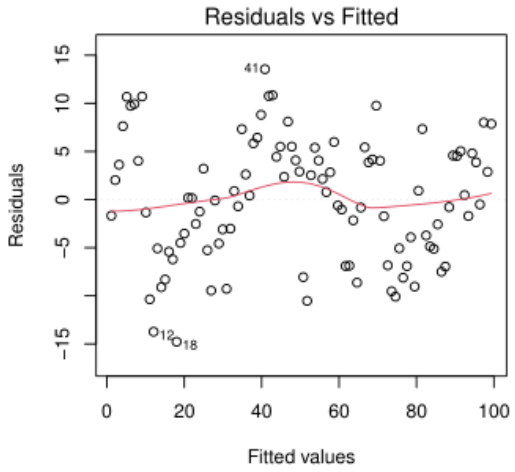


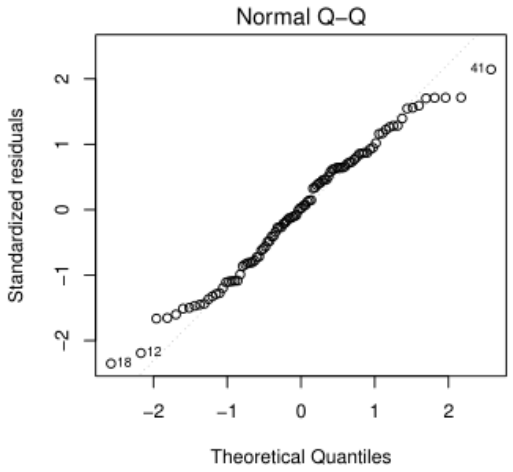


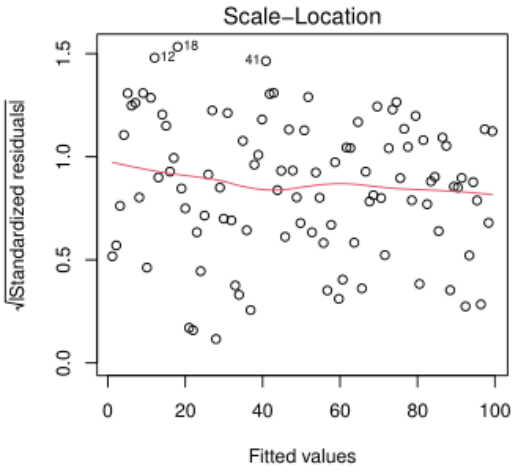
Residual plots for (x,y5)

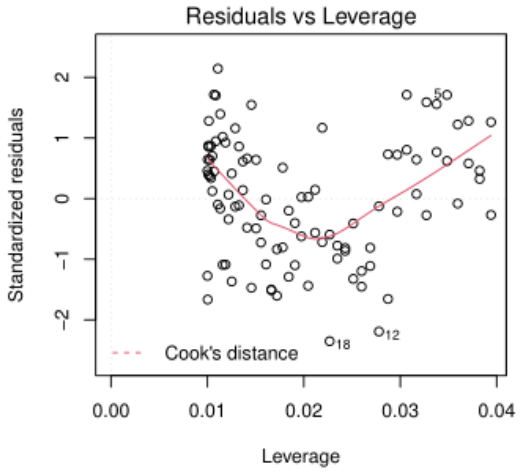


Residual plots for (x,y6)

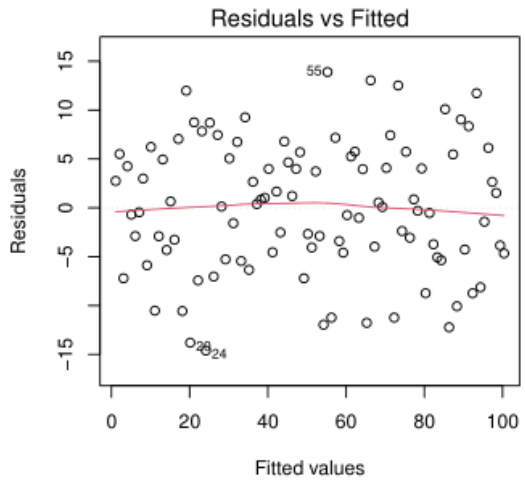


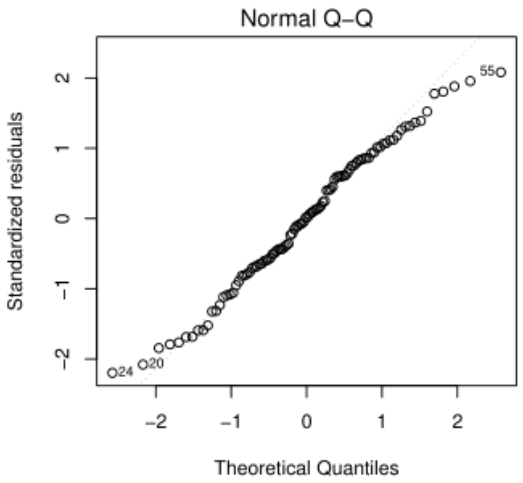


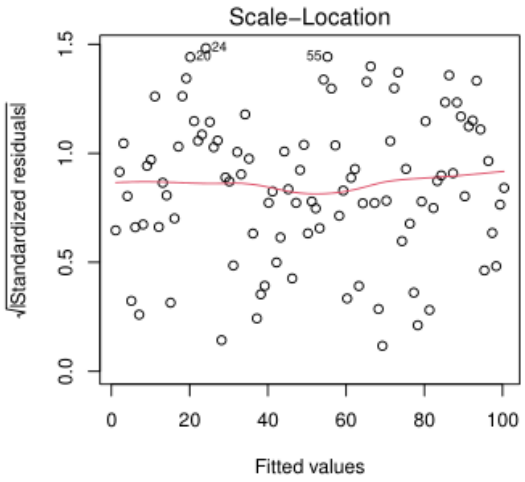


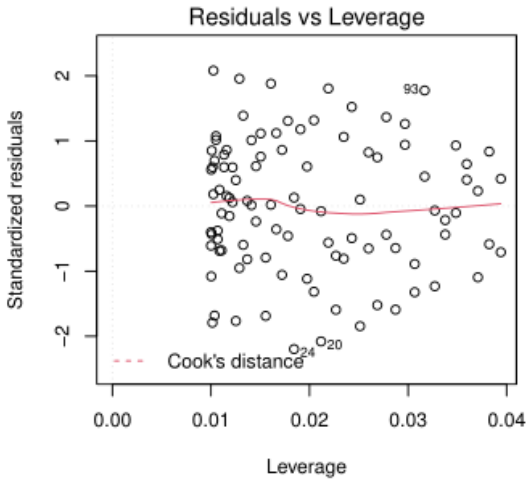


Residual plots for (x,y7)

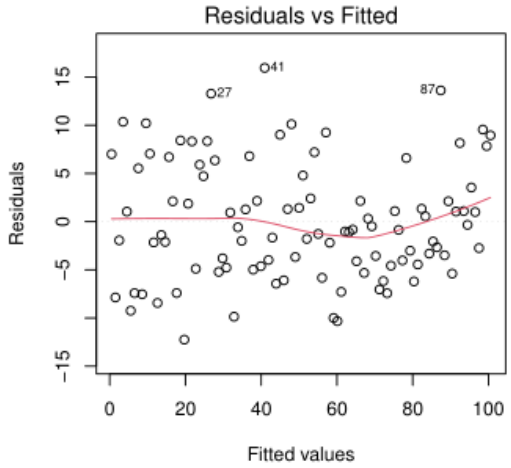


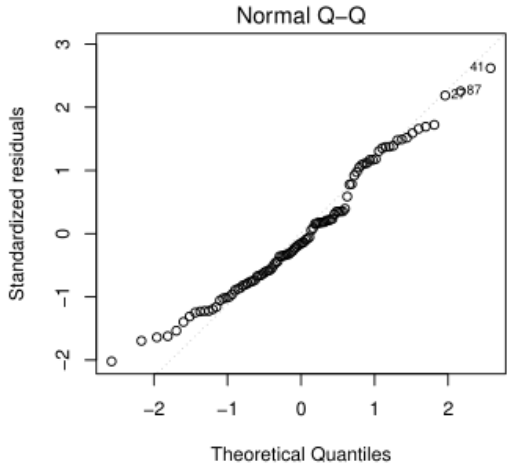


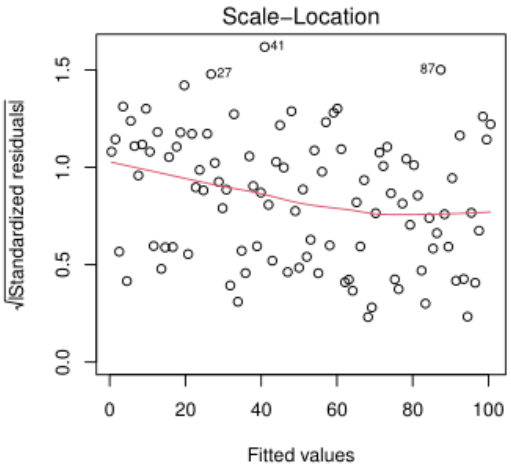


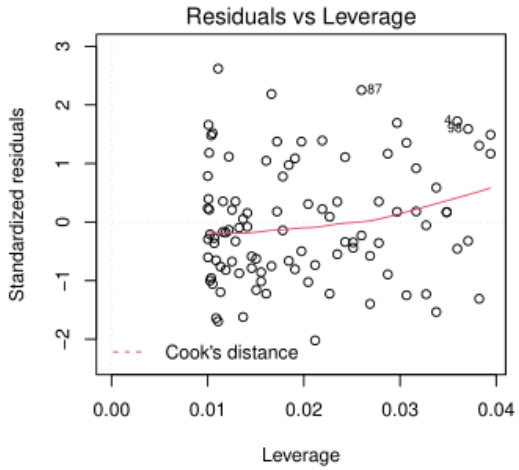


Residual plots for (x,y8)





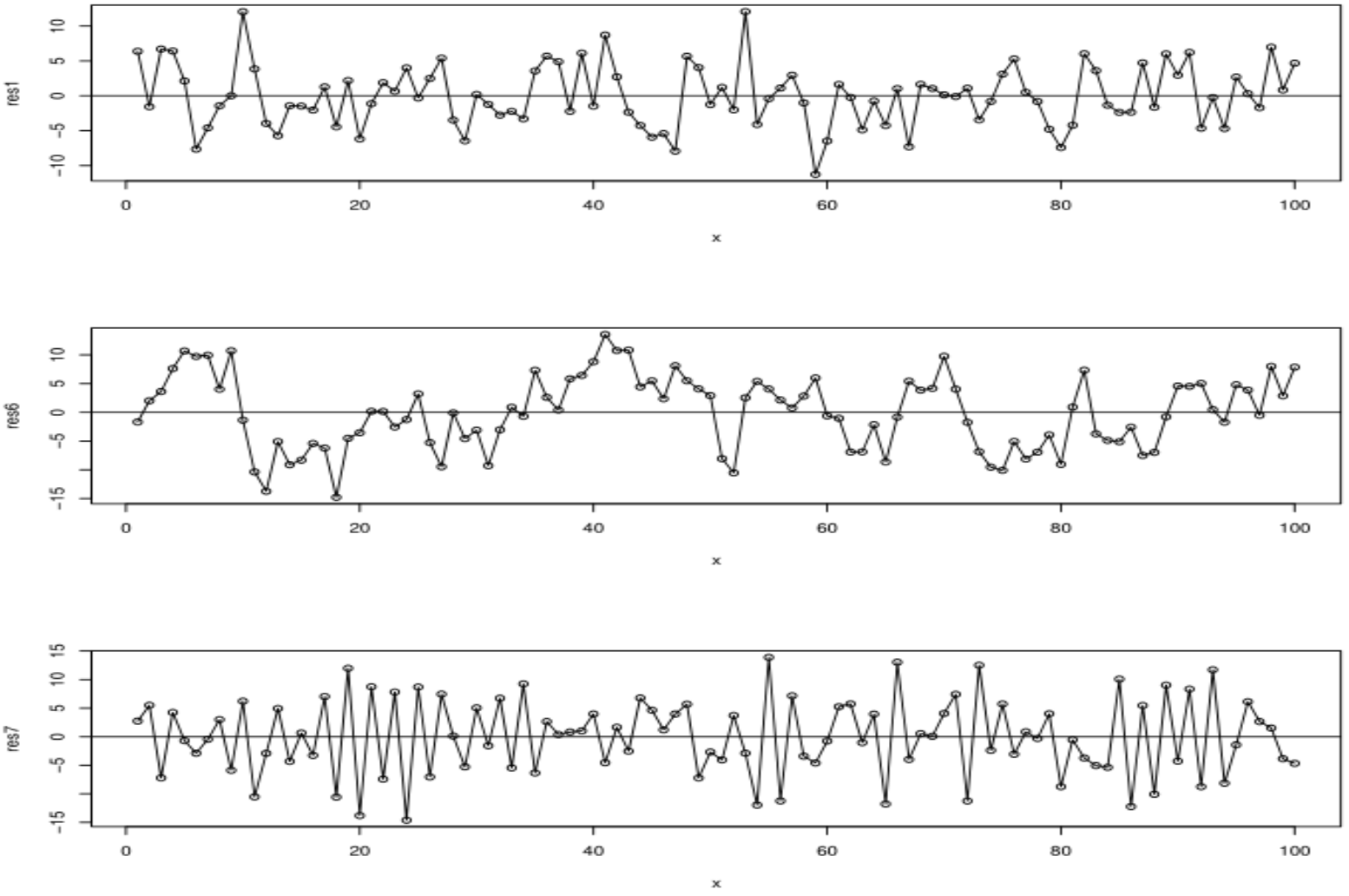




Summary of findings

* (x,y1): shows how the plots should be when all assumptions are reasonable.
* (x,y2): curvature in Residual vs Fitted=>nonlinearity between x and y2
* (x,y3): megaphone pattern in Residual vs Fitted, increasing trend in Scale-Location plot=>non constant error variance
* (x,y4): presence of outlier
* (x,y5): Normal Q-Q plot deviates from straight line => nonnormal error
* (x,y6), (x,y7), (x,y8): anything wrong? Need a further look

Plot of residuals against time



Summary of findings

* Plot of res1 vs x for (x,y1): signs of residuals change randomly across the zero line.
* Plot of res6 vs x for (x,y6): signs of residuals change much more slowly across the zero line => positive correlation among consecutive errors.
* Plot of res7 vs x for (x,y7): signs of residuals change much more quickly across the zero line => negative correlation among consecutive errors.
* Negative correlation may be hard to see by eyes.

When does correlation occur?

* Repeated measurements on one unit
* Business and economic data observed over time tend to be positively correlated
* Business data such as monthly sales or quarterly revenues may exhibit seasonal pattern
* Survey taken from people living in the same household
* Housing prices in the same residential area

Why do we care about correlation?

* Effect on inference
  * Estimated regression coefficients unbiased, but no longer has minimum variance
  * MSE may seriously underestimate the variance of the error terms
  * SE for estimated coefficients may seriously underestimate the true ones
  * Confidence intervals and tests no longer appropriate
* Effect on prediction
  * Correlation can help achieve more accurate predictions

How do we test it? (not required)

* A first-order autoregressive model

$$Y_t = \beta_0 + \beta_1 X_t + \epsilon_t$$

  * $\epsilon_t = \rho\epsilon_{t-1} + u_t$
  * $u_t \sim Normal(0, \sigma^2)$
* The simple linear regression model assumes $\rho = 0$.
* How about we just test
  * $H_0: \rho = 0 \ \ vs \ \ H_a: \rho > 0 \ \ for \ postive \ correlation$
  * $H_0: \rho = 0 \ \ vs \ \ H_a: \rho < 0 \ \ for \ negative \ correlation$
  * $H_0: \rho = 0 \ \ vs \ \ H_a: \rho \neq 0 \ \ for \ any \ correlation$

The Durbin-Watson Test

* The Durbin-Watson test statistic:

$$D = \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2}$$

* The test statistic value is between 0 and 4.
* It can be shown that when $\rho = 0$, the Durbin-Watson test statistic should be around 2.
* What happens under strong positive correlation? Say $\rho = 1$?
* What happens under strong negative correlation? Say $\rho = -1$?

Test for presence of correlation

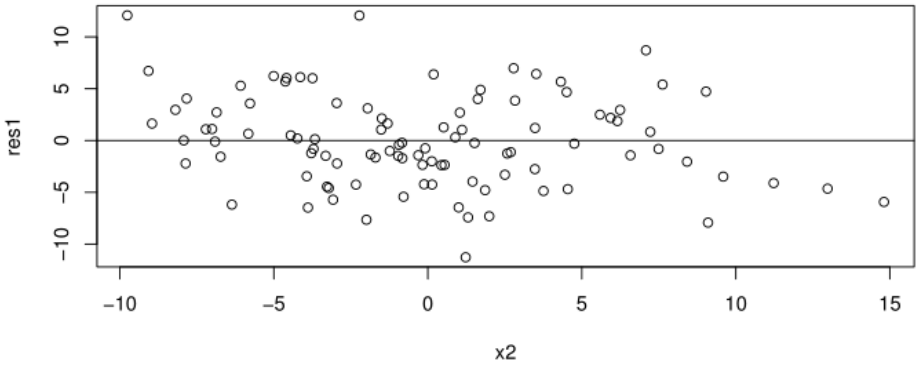
```
> library(car)
> m1<-lm(y1~x)
> durbinWatsonTest(m1)
 lag Autocorrelation D-W Statistic p-value
   1     -0.03601692       2.056127   0.862
 Alternative hypothesis: rho != 0
> m6<-lm(y6~x)
> durbinWatsonTest(m6)
 lag Autocorrelation D-W Statistic p-value
   1       0.7278706     0.5409363       0
 Alternative hypothesis: rho != 0
> m7<-lm(y7~x)
> durbinWatsonTest(m7)
 lag Autocorrelation D-W Statistic p-value
   1      -0.7042901       3.386276       0
 Alternative hypothesis: rho != 0
```

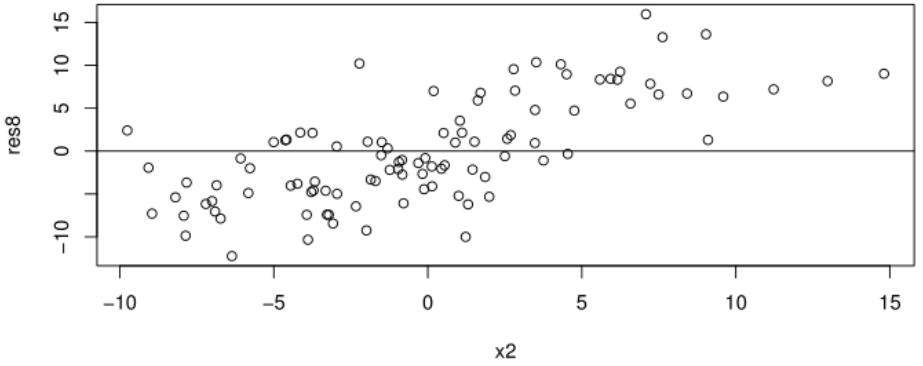
Test for positive/negative correlation

```
> durbinWatsonTest(m6,alternative="positive")
 lag Autocorrelation D-W Statistic p-value
   1       0.7278706     0.5409363       0
 Alternative hypothesis: rho > 0
> durbinWatsonTest(m6,alternative="negative")
 lag Autocorrelation D-W Statistic p-value
   1       0.7278706     0.5409363       1
 Alternative hypothesis: rho < 0
> durbinWatsonTest(m7,alternative="positive")
 lag Autocorrelation D-W Statistic p-value
   1      -0.7042901      3.386276       1
 Alternative hypothesis: rho > 0
> durbinWatsonTest(m7,alternative="negative")
 lag Autocorrelation D-W Statistic p-value
   1      -0.7042901      3.386276       0
 Alternative hypothesis: rho < 0
```

Plot of residuals against omitted predictor

* x2 is an omitted predictor
* The plot of res8 vs x2 where res8 is from (x,y8) shows an increasing pattern. However, this is not true for res1 vs x2 where res1 is from (x,y1).
* Thus, x2 may be included in the model for (x,y8) but not for (x,y1).



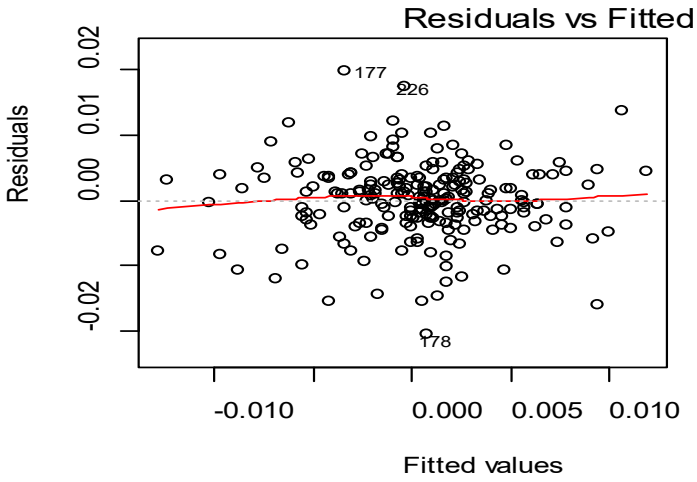


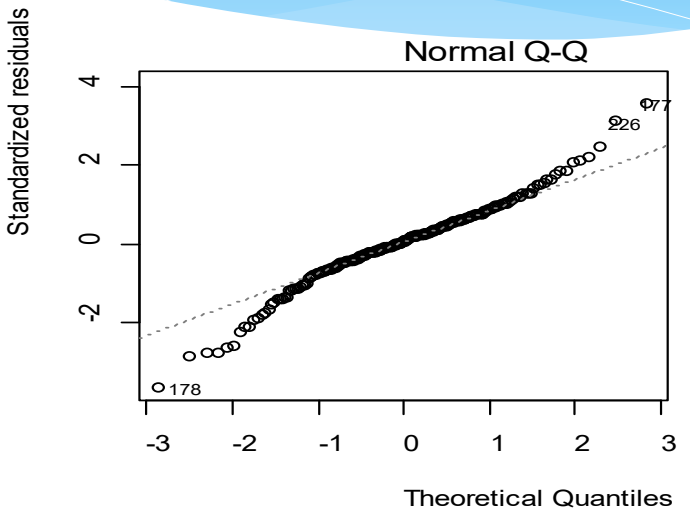
Example: How are different stock indices related?

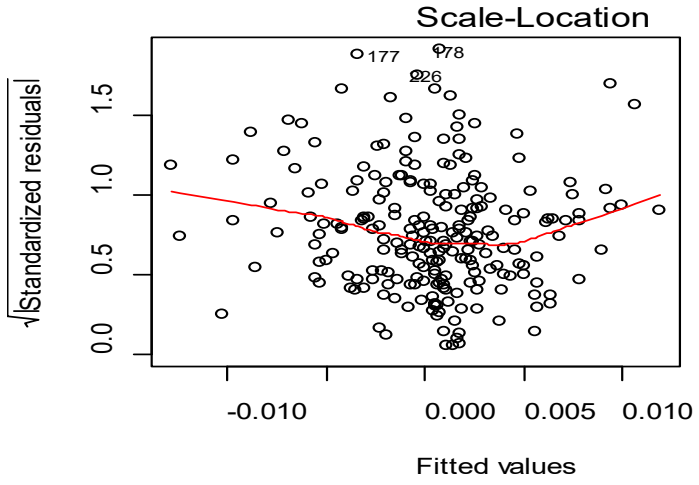
Use appropriate plot(s) to comment on the following questions

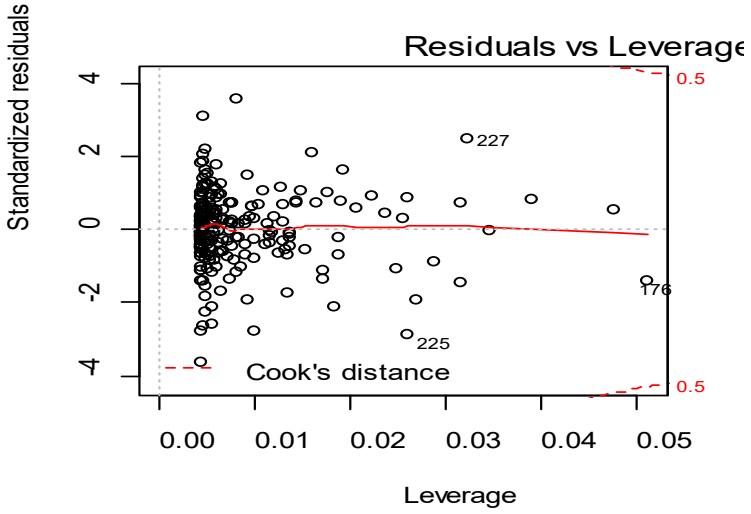
- if the regression model is linear (use Residuals vs Fitted, together with scatter plot)
- if the error terms have constant variance (use Residuals vs Fitted and/or Scale-Location)
- if the error terms are independent (plot residuals against time)
- if there are outliers present (use Residuals vs Fitted and/or Residuals vs Leverage)
- if the error terms are normally distributed (use Normal Q-Q plot)
- if either UKX or SSE should be included in the model(use Residuals vs omitted predictor)

R default residual plots

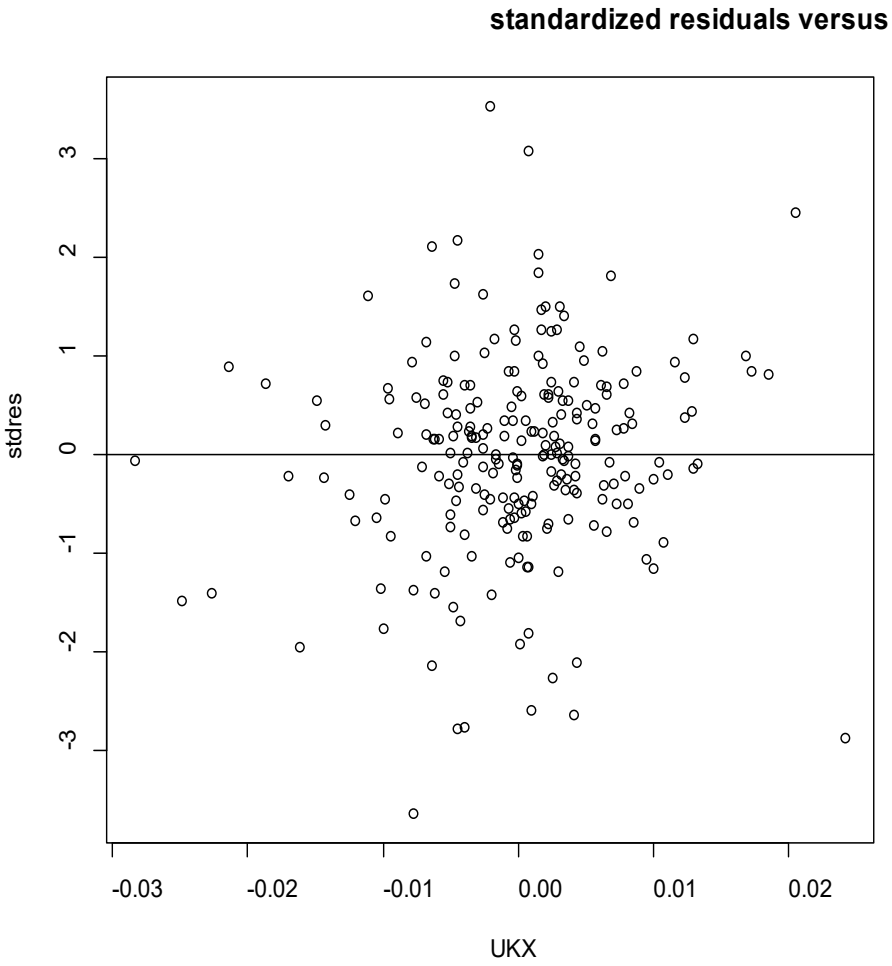


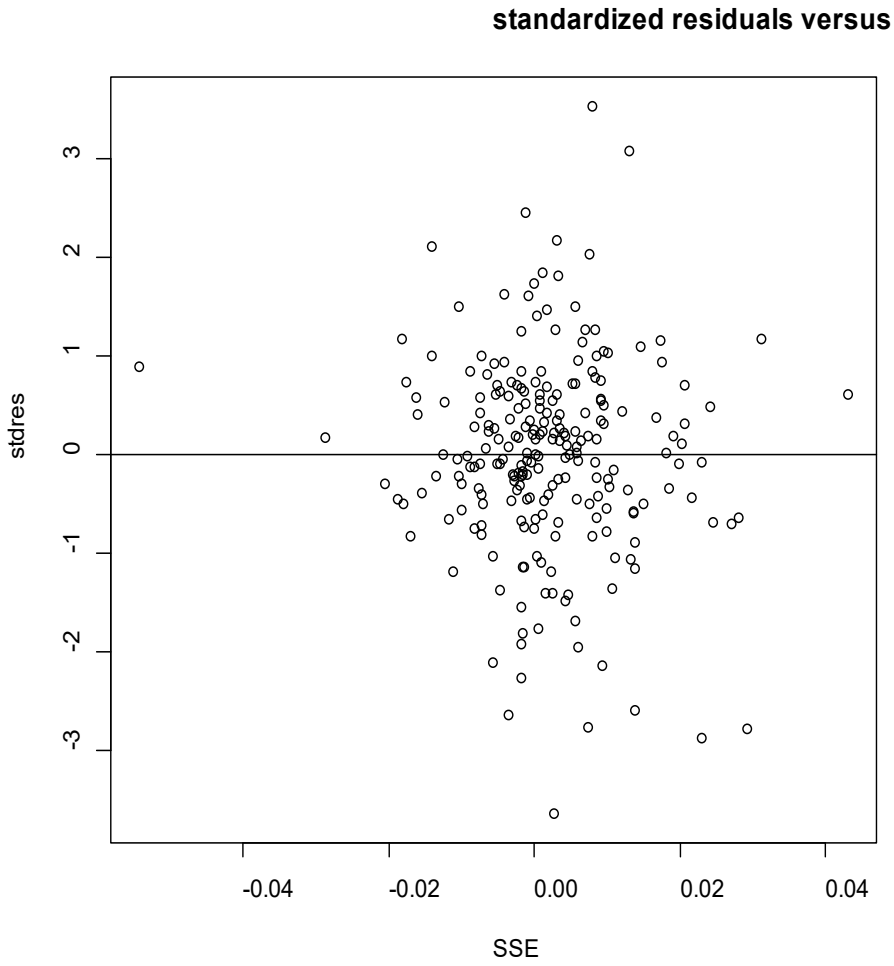




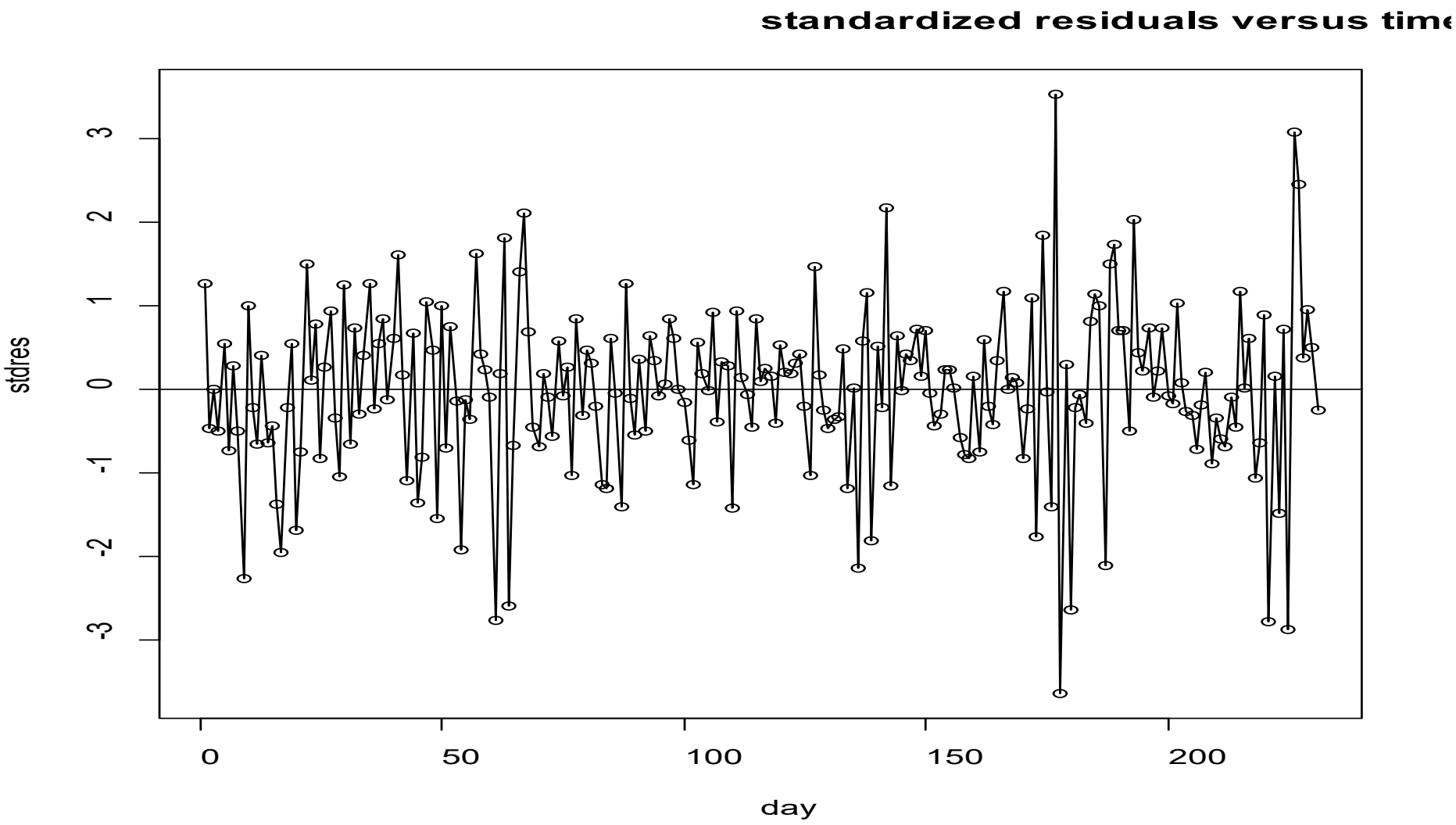


Standardized residuals vs omitted predictor





Standardized residuals vs time



Durbin-Watson Test

* The Durbin-Watson test results are given below.

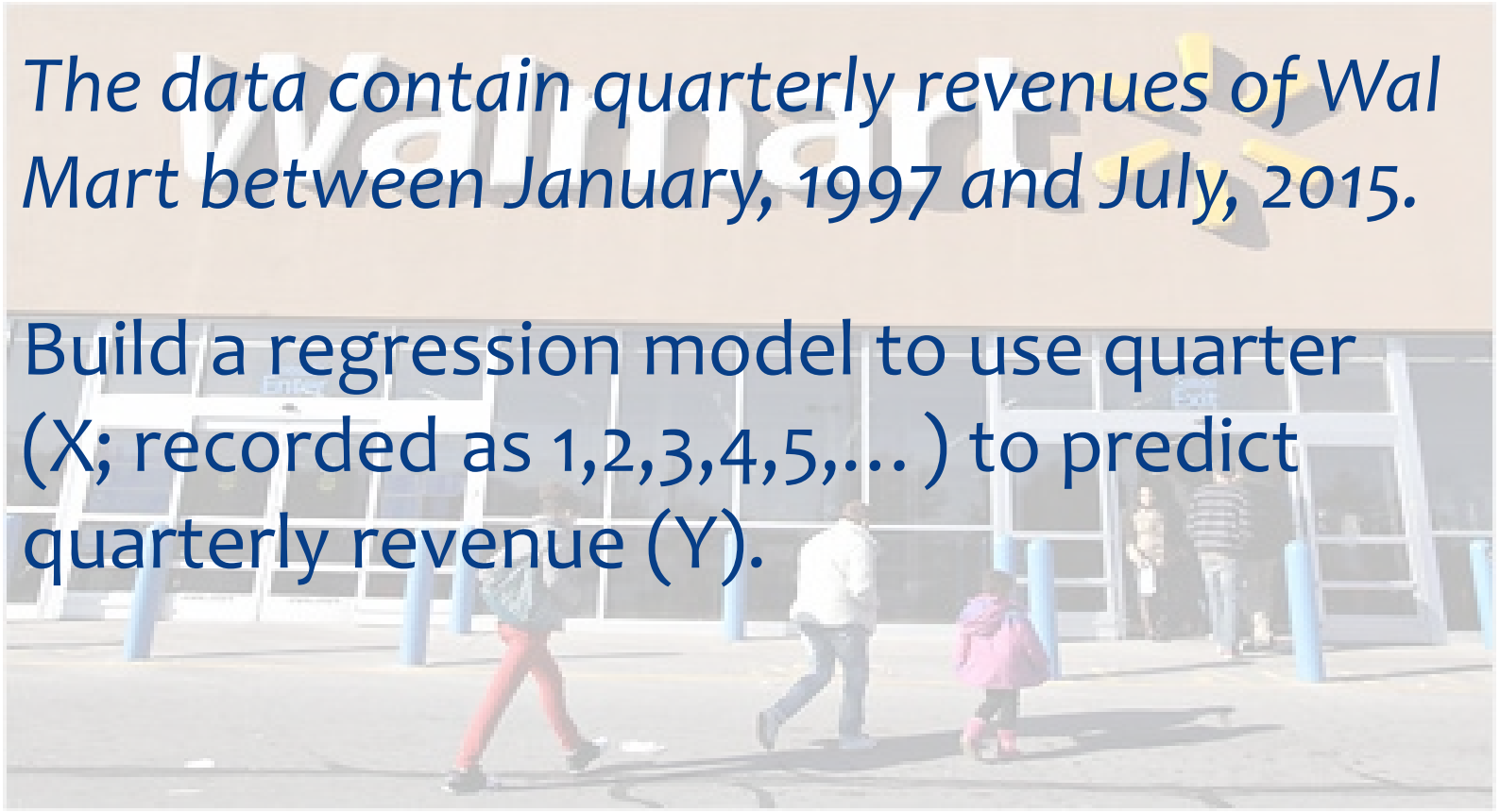
```
> m<-lm(DJ~DAX)
> durbinWatsonTest(m)
 lag Autocorrelation D-W Statistic p-value
   1       -0.2052347       2.403307   0.004
 Alternative hypothesis: rho != 0
> durbinWatsonTest(m,alternative="negative")
 lag Autocorrelation D-W Statistic p-value
   1       -0.2052347       2.403307   0.001
 Alternative hypothesis: rho < 0
> durbinWatsonTest(m,alternative="positive")
 lag Autocorrelation D-W Statistic p-value
   1       -0.2052347       2.403307       1
 Alternative hypothesis: rho > 0
```

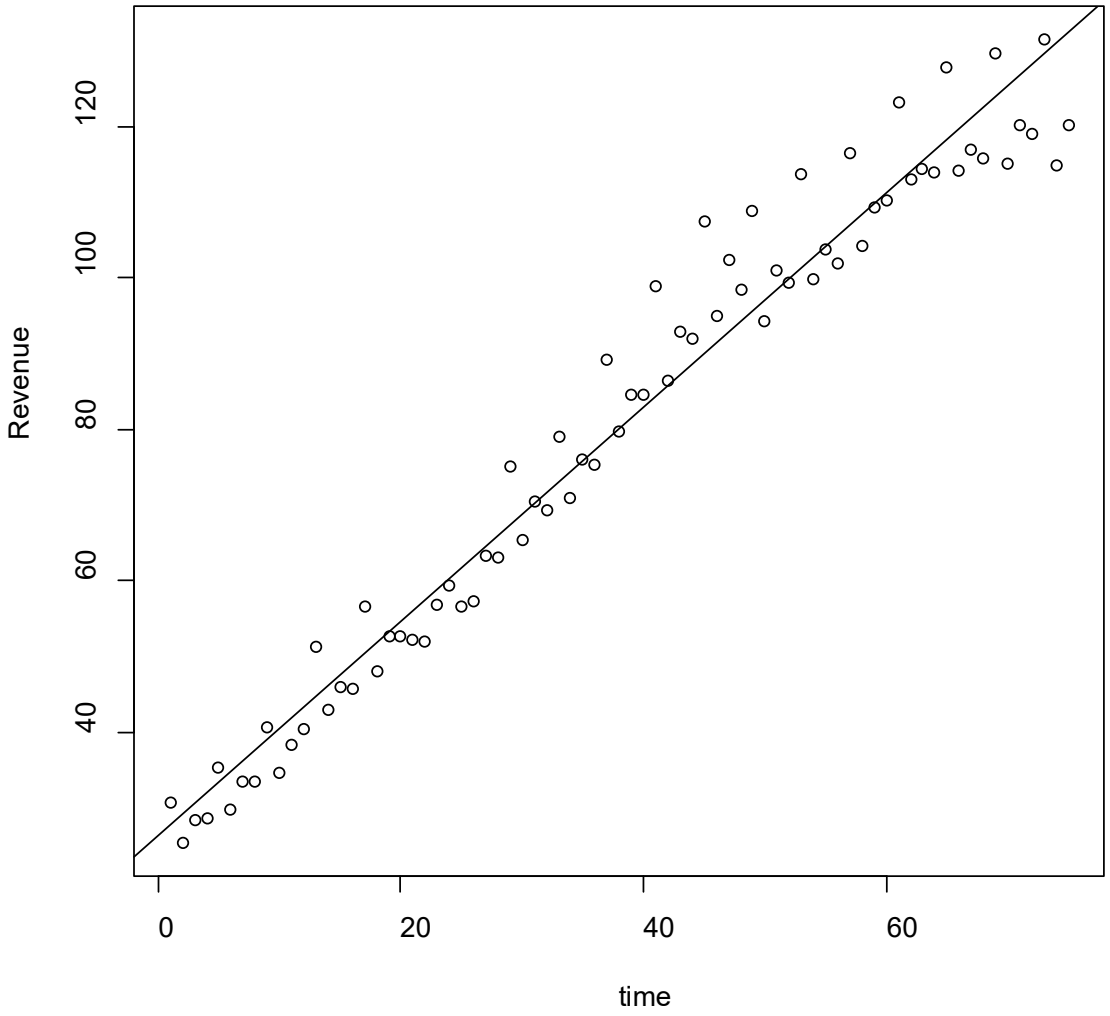
* Conclusion:

Example: How to model and forecast WalMart's quarterly revenue?

The data contain quarterly revenues of Wal Mart between January, 1997 and July, 2015.

Build a regression model to use quarter (X; recorded as 1,2,3,4,5,…) to predict quarterly revenue (Y).

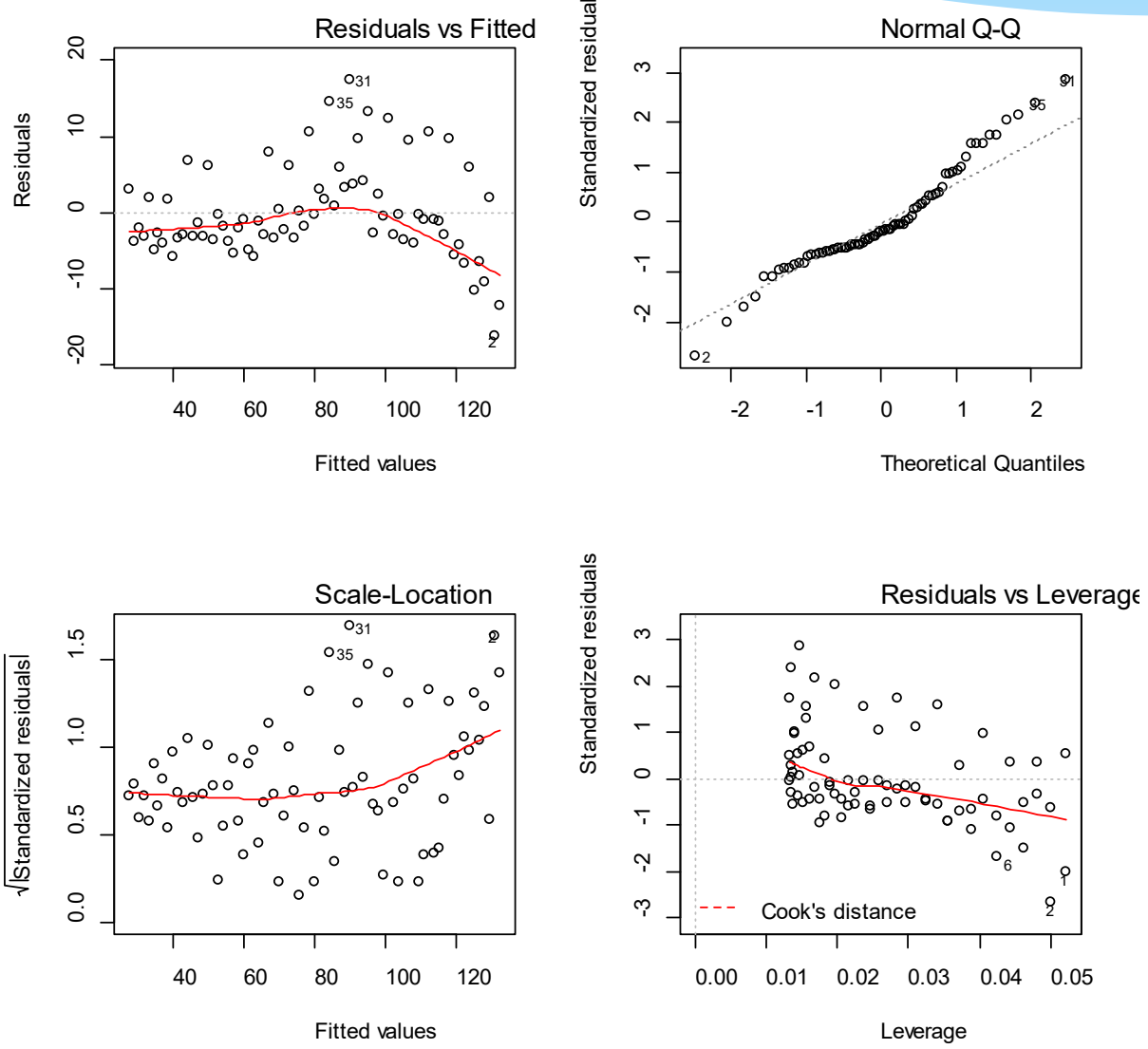


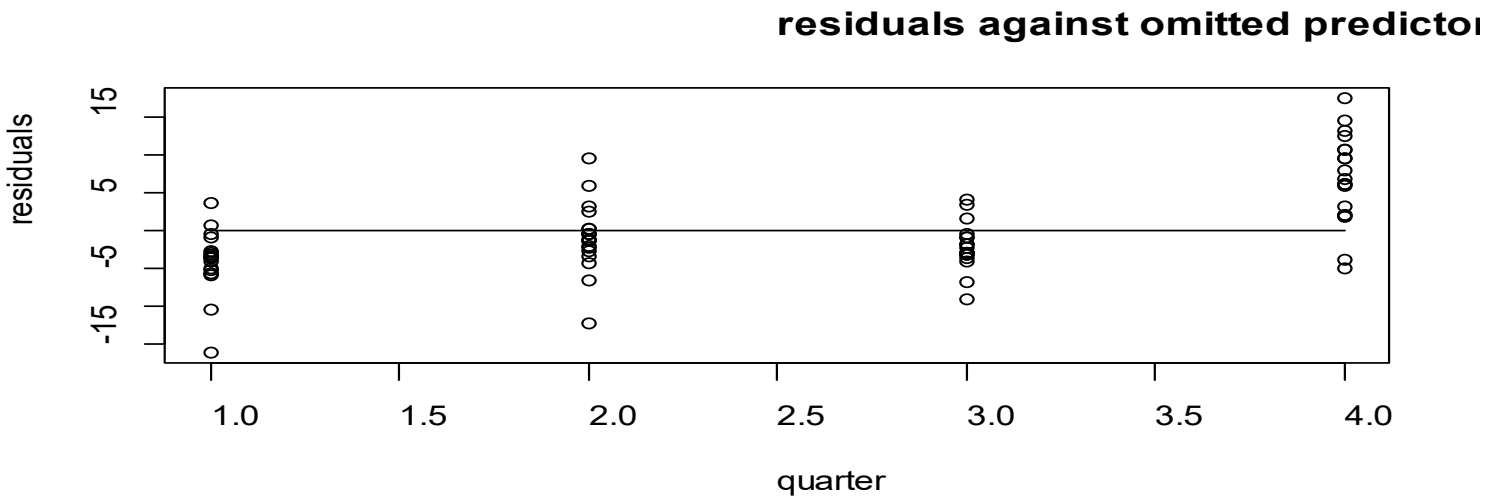

Revenue
40
60
80
100
120
0
20
40
60
time

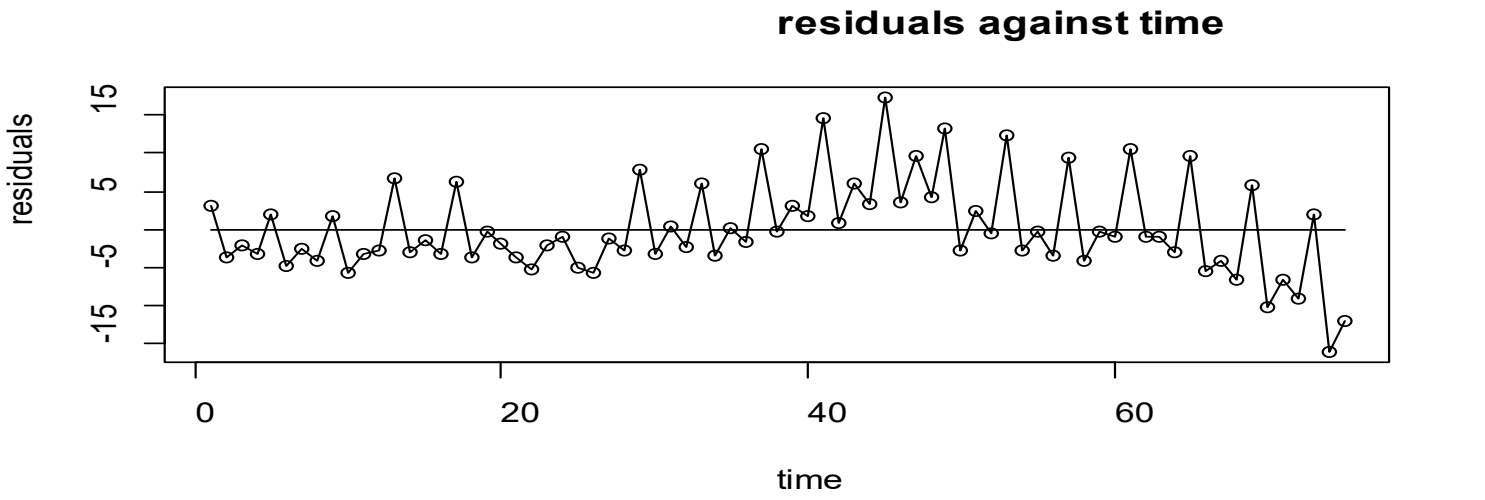
Example: How to model and forecast WalMart's quarterly revenue?

Use appropriate plot(s) to comment on the following questions

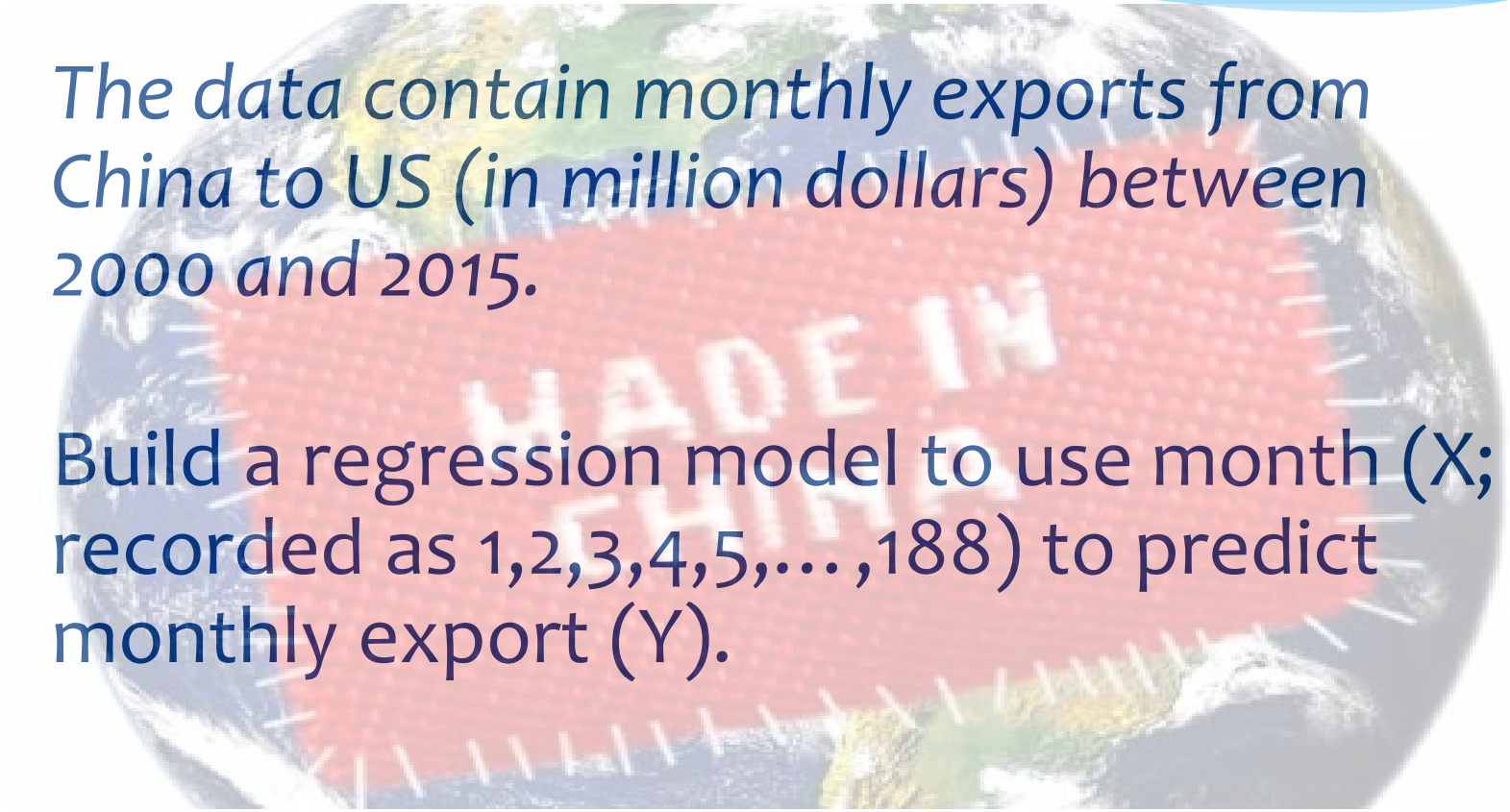
- if the regression model is linear (use Residuals vs Fitted, together with scatter plot)
- if the error terms have constant variance (use Residuals vs Fitted and/or Scale-Location)
- if the error terms are independent (plot residuals against time)
- if there are outliers present (use Residuals vs Fitted and/or Residuals vs Leverage)
- if the error terms are normally distributed (use Normal Q-Q plot)
- if quarter in a year (i.e., 1, 2, 3, 4) should be included in the model (use Residuals vs omitted predictor)


Residuals vs Fitted
Residuals
Fitted values
Normal Q-Q
Standardized residuals
Theoretical Quantiles
Scale-Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage


residuals against omitted predictor
residuals
15
5
-5
-15
1.0
1.5
2.0
2.5
3.0
3.5
4.0
quarter

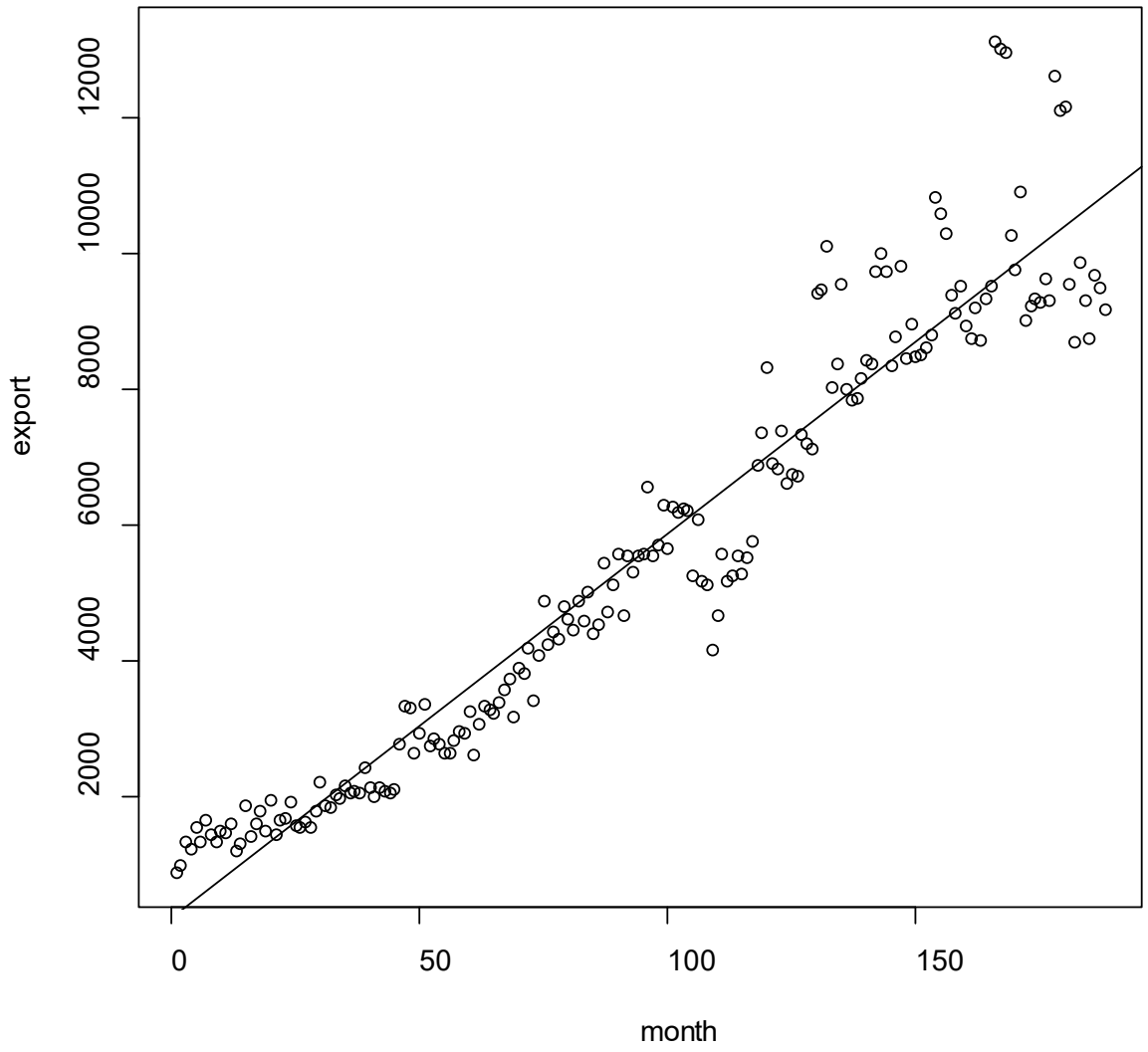

residuals against time
residuals
15
5
-5
-15
0
20
40
60
time

Example: How to model monthly export from China to the US?



The data contain monthly exports from China to US (in million dollars) between 2000 and 2015.

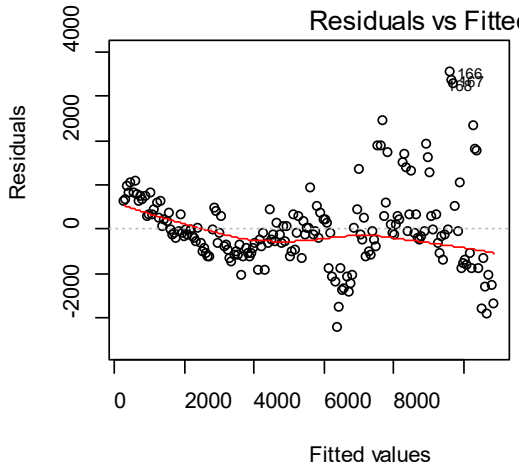
Build a regression model to use month (X; recorded as 1,2,3,4,5,…,188) to predict monthly export (Y).

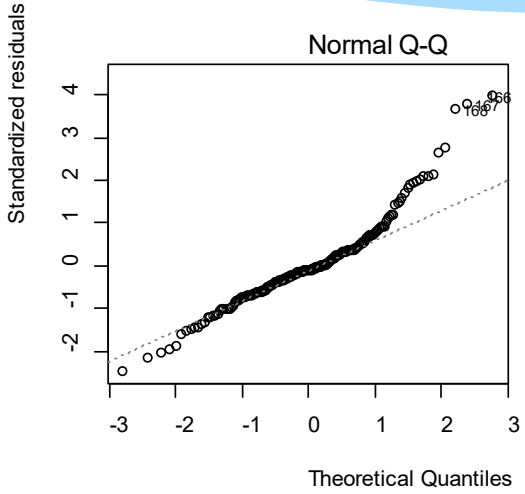

export
2000
4000
6000
8000
10000
12000
0
50
100
150
month

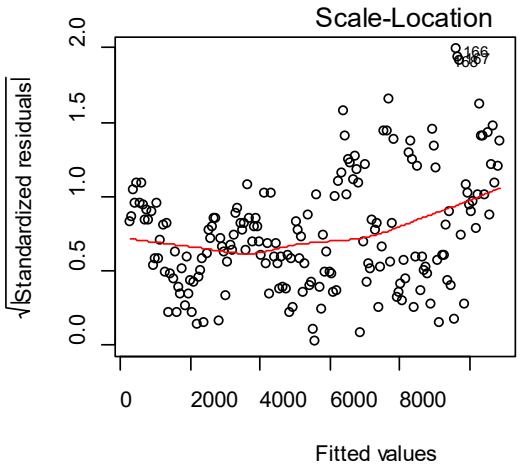
Example: How to model monthly export from China to the US?

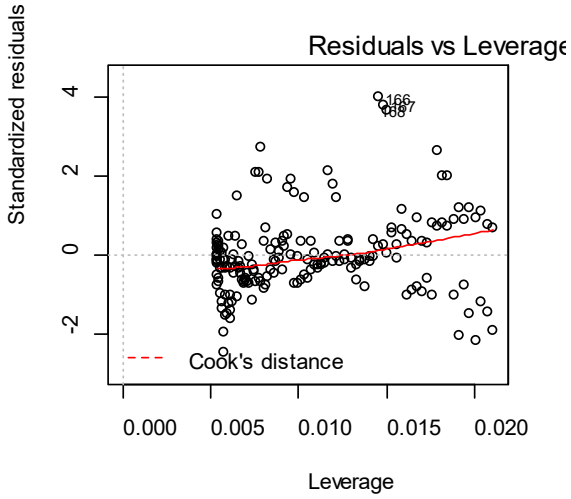
Use appropriate plot(s) to comment on the following questions

- if the regression model is linear (use Residuals vs Fitted, together with scatter plot)
- if the error terms have constant variance (use Residuals vs Fitted and/or Scale-Location)
- if there are outliers present (use Residuals vs Fitted and/or Residuals vs Leverage)
- if the error terms are normally distributed (use Normal Q-Q plot)


Residuals vs Fitted
Residuals
Fitted values
4000
2000
0
-2000
0
2000
4000
6000
8000


Normal Q-Q
Standardized residuals
Theoretical Quantiles
4
3
2
1
0
-1
-2
-3
-2
-1
0
1
2
3


Scale-Location
√|Standardized residuals|
Fitted values
2.0
1.5
1.0
0.5
0.0
0
2000
4000
6000
8000


Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
4
2
0
-2
0.000
0.005
0.010
0.015
0.020

Some remarks

* Violation of normality is generally not a serious problem for inference, especially when you have large data. However, the prediction interval will be invalid if your data are not normal.
* Violation of equal variance assumption will cause problem to all your analyses so you need to be careful.
* What may appear to be, *for example, non-normality of the error terms may, indeed, be nonlinearity.* That is, it is easy to misdiagnose a problem. So, you must make sure that your analysis is thorough.
* Make sure you don't see patterns that don't exist.

So you have fitted a model, then what?

More questions to ask:

* Does the model help (or do we need to bother)?
* What can we say about the regression coefficients?
* Suppose that DAX change by 1%, how should we expect the DJ to change on average? How do we predict the actual change of DJ on a particular day?
* Does the fitted model fit the data well? Are any model assumptions being violated?
* If there is a problem, what do we do?



Dr. Guan

Remedial Procedures

How to deal with outliers?

Investigation of the point in the practical setting of the problem must be used to determine if the point is an actual outlier. The "oddness" of the observation could be due to

- *measurement error*
- *recording error*
- *other extraneous variables impacting the observation*
- etc.

So, what do you do with it?

* Use your best judgement. It is generally best to err on the conservative side and leave the point in the sample unless you have a justifiable reason to believe it should be removed.

* There are regression procedures that are less affected the presence of outliers.

What if the relationship between X and Y appears nonlinear?

* *Example:* $Y_i = \beta_0 + \beta_1 X_i + \beta_2 {X_i}^2 + \varepsilon_i$

A good approach is to fit a multiple linear regression model with a quadratic term included. This will be discussed later.

* *In general, we are not interested in the theoretical form of the model, but simply wish to make predictions. In this case, an arithmetic transformation of X or Y or both may yield a relationship that is nearly linear.*

What if the variance is nonconstant?

* *Generally you should try to **transform Y**.*
* *The **Box-Cox transformations** is useful to select the best (power) transformation.*

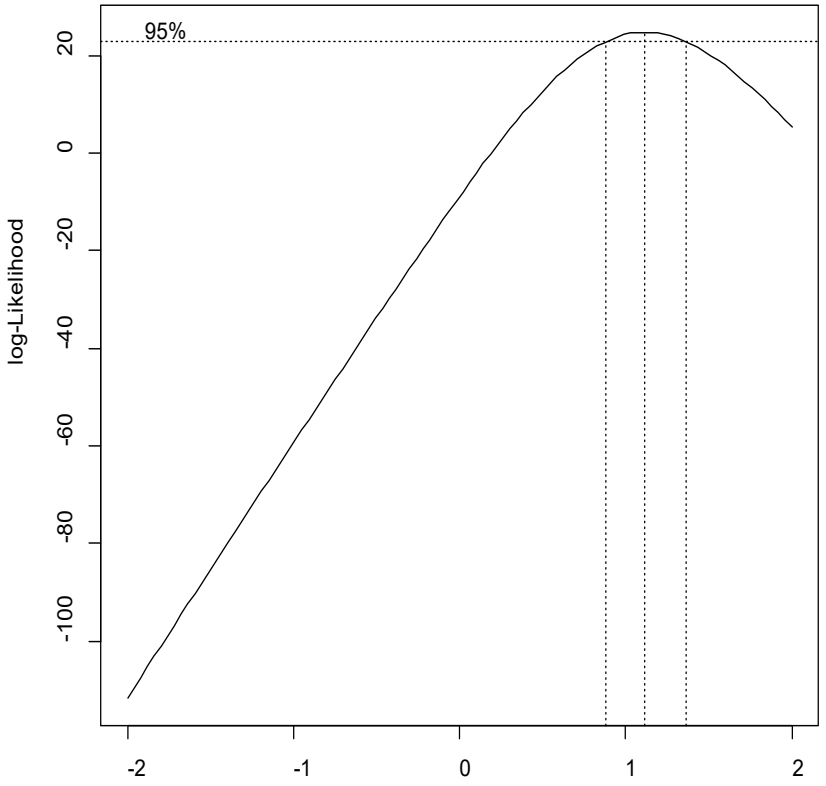
$$Y_i^* = \frac{{Y_i}^\lambda - 1}{\lambda} \; or \; just \; Y_i^* = Y_i^\lambda \; if \; \lambda \neq 0,$$
$$Y_i^* = \log(Y_i) \; if \; \lambda = 0.$$

* *We will treat Y_i^* as the new response.*
* *Typically we use some that are easier to explain. For example, rather than using $\lambda = 0.45$, we may use $\lambda = 0.5$, i.e., a square root transformation*
* *For many economic data, **a logarithm transformation** often works well.*

Example: How to model and forecast WalMart's quarterly revenue?

Box-Cox transformation:

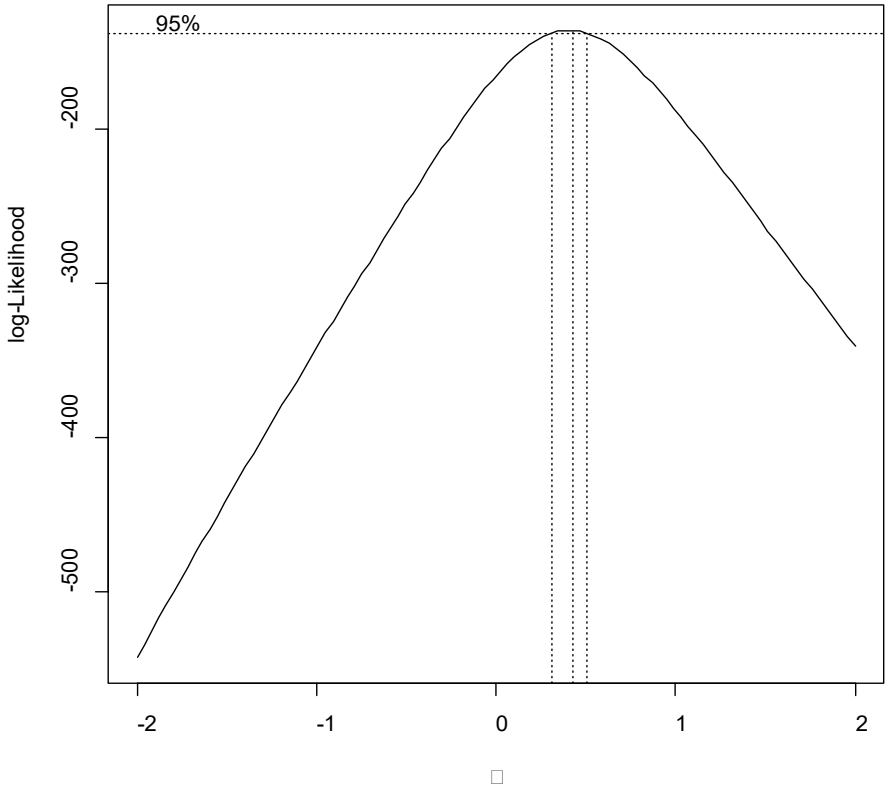
* The middle vertical line shows the best estimated λ.
* The other two vertical lines show 95% confidence interval for λ.
* No transformation is needed since 1 is inside the interval.



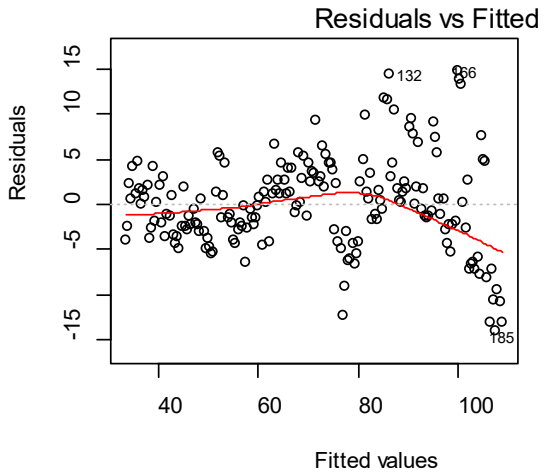
Example: How to model monthly export from China to the US?

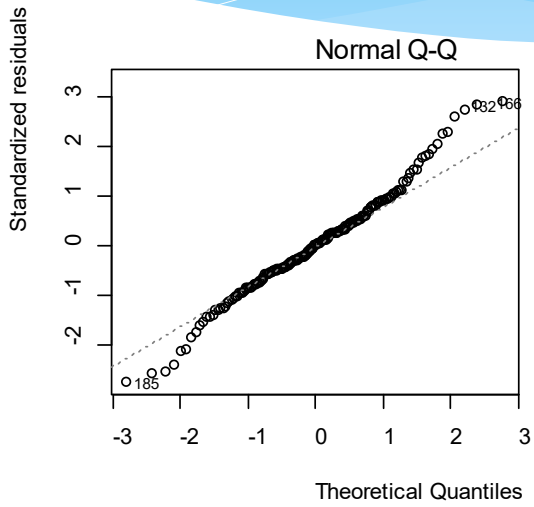
Box-Cox transformation:

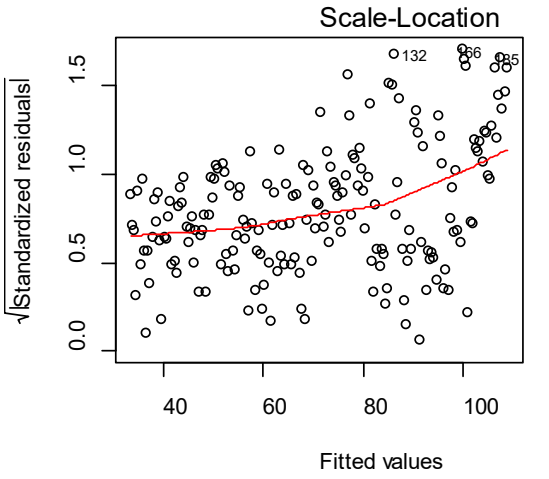
* The best estimated λ is 0.4242.
* 0.5 is within the 95% confidence interval for λ.
* A square root transformation may be appropriate.

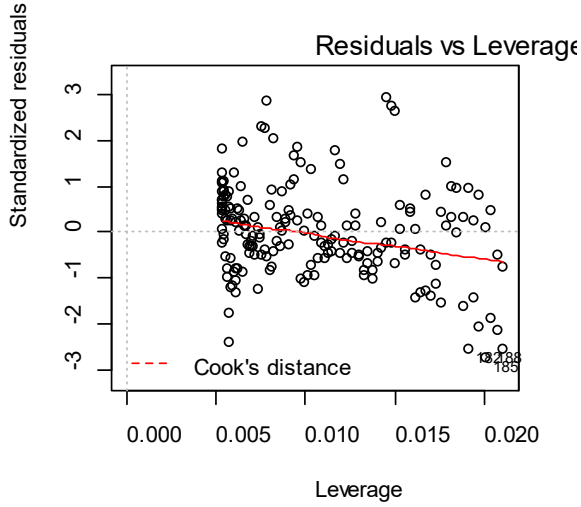


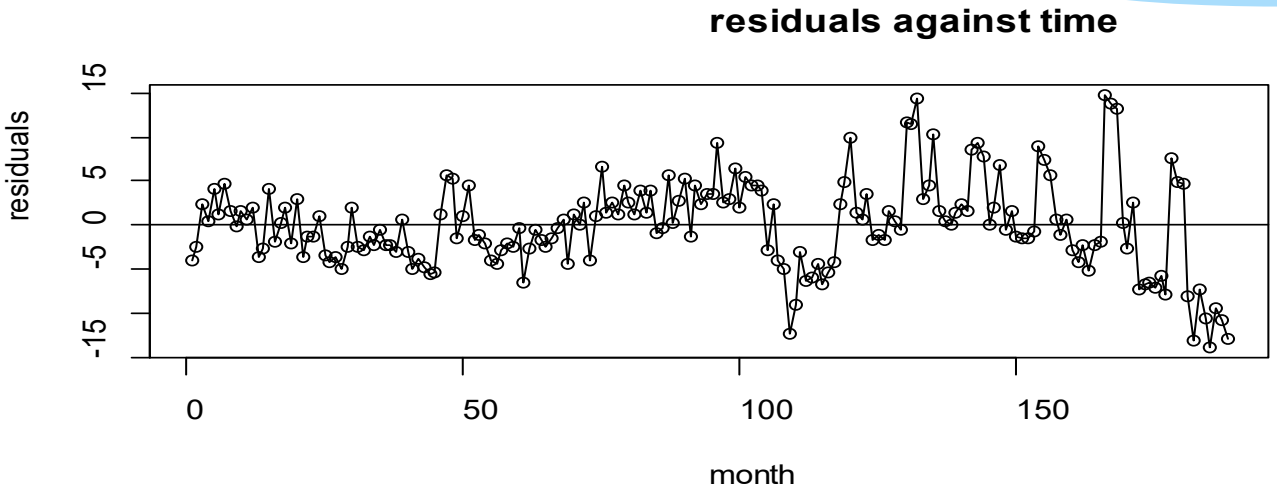
Residual plots after the transformation

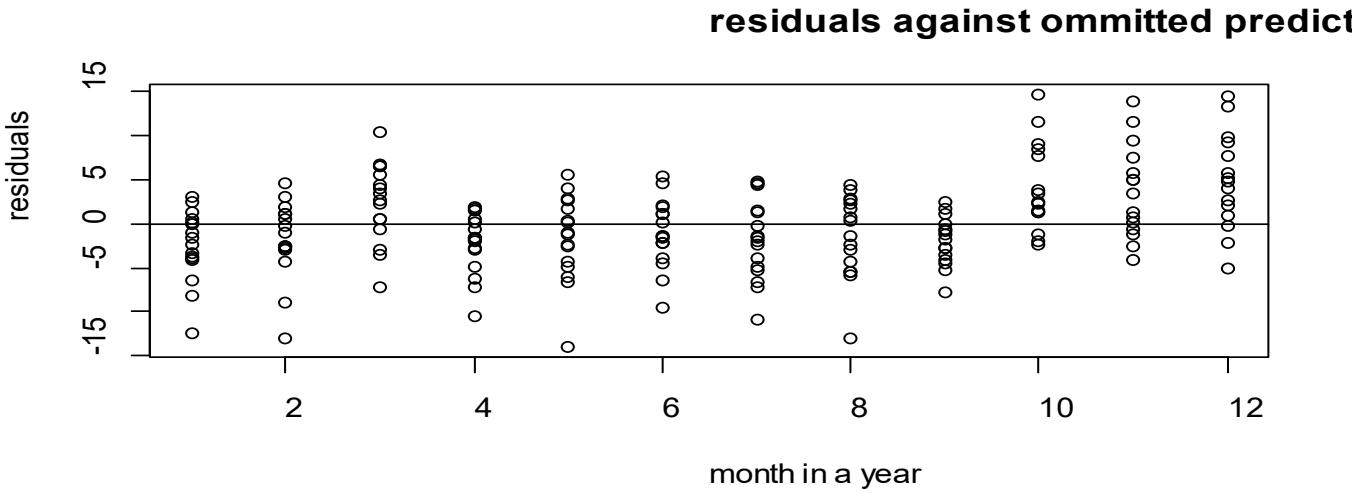









residuals against time
residuals
15
5
0
-5
-15
0
50
100
150
month


residuals against ommitted predict
residuals
15
5
0
-5
-15
2
4
6
8
10
12
month in a year

```
> durbinWatsonTest(m_bc)
 lag Autocorrelation D-W Statistic p-value
   1       0.6197906     0.7225846       0
 Alternative hypothesis: rho != 0
> durbinWatsonTest(m_bc,alternative="positive")
 lag Autocorrelation D-W Statistic p-value
   1       0.6197906     0.7225846       0
 Alternative hypothesis: rho > 0
> durbinWatsonTest(m_bc,alternative="negative")
 lag Autocorrelation D-W Statistic p-value
   1       0.6197906     0.7225846       1
 Alternative hypothesis: rho < 0
```